

Navigating Through Fear and Greed: The Experience-Driven Disposition Effect *

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Abstract

Using transaction-level data on 189,530 Chinese retail investors over the 2013–2016 boom and bust, we measure experience as counts of large realized gains and losses and examine how these counts relate to the disposition effect, the tendency to sell winners and hold losers. A two-standard-deviation increase in accumulated large losses raises the propensity to sell winners rather than losers by about 23% of the baseline effect, while an equivalent increase in gains lowers it by about 21%. Under composition-matched simulations, experience-independent selling rules do not reproduce this asymmetry. To explain both the baseline effect and this asymmetry, we build a memory-based recall model in which similarity-weighted retrieval of past outcomes guides selling. A single retrieval channel reproduces all three patterns without shifting preferences.

Keywords: Retail trading, disposition effect, investor experience, memory, cued recall

JEL Codes: D14, D91, G11, G14, G41

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1 Introduction

Between July 2013 and February 2016, the Chinese stock market more than doubled and then shed over 40% of its value in less than a year. We use this boom and bust, together with complete within-brokerage trading records for 189,530 retail investors at one of China’s largest brokerages, to ask how the gains and losses an investor accumulates reshape the disposition effect, the tendency to sell winners and hold losers (Shefrin and Statman, 1985; Odean, 1998). Field studies document a weaker disposition effect among more experienced investors and interpret this pattern as evidence of increasing sophistication (Feng and Seasholes, 2005; Seru, Shumway and Stoffman, 2010). We find that the disposition effect weakens as realized gains accumulate but strengthens as realized losses accumulate. Investors with more large realized losses exhibit a larger gap between the probabilities of selling winners and losers.

The asymmetry is economically large. We measure experience as an investor’s cumulative count of realized outcomes, trades closed more than $\pm 5\%$ from the average purchase price, and estimate how the propensity to sell a winner rather than a loser moves with the counts of past gains and past losses. A two-standard-deviation increase in accumulated losses raises that propensity by 0.57 percentage points, about 23% of the 2.48-percentage-point baseline disposition effect; a two-standard-deviation increase in gains lowers it by 0.51 points, about 21%. Both estimates are precise ($t = 3.3$ and -3.3) and hold under investor, stock, and time fixed effects, with standard errors clustered on all three dimensions.

To organize these facts, we develop a memory-based model of the selling decision that adapts the temporal-context framework of Howard and Kahana (2002) to an investor’s portfolio. The investor stores each holding as an episode, a stock paired with the outcome she experiences while owning it; when she decides what to sell, the cue “stocks I own” retrieves these episodes in proportion to their similarity and frequency, each competing with the others for recall. Large moves occur almost only in stocks and, for stocks, are positively skewed (Oh and Wachter, 2022), so a large gain is what most readily comes to mind, and the investor sells the winner that does. This produces the baseline disposition effect. Realizing a gain reclassifies the outcome from owned to sold, weakening its association with ownership and reducing the bias. Realizing a loss strengthens the association between ownership and

remaining gains, increasing the bias. One retrieval channel thus generates the baseline effect, its decay after gains, and its amplification after losses, with no change in preferences.

Because experience is constructed from prior sales, we examine whether the interactions arise mechanically from changes in portfolio composition. Under the calibrated selling rules, the mean mechanical loss-side interaction is negative or close to zero, unlike the positive interaction in the data (Section 5). The effect also survives a range of competing explanations, from unobserved skill and rebalancing to return-extrapolation and overtrading accounts.

Three contributions follow. First, the content of experience, not its quantity, governs the disposition effect. A field literature proxies experience by the number of trades and reads the fading bias as growing sophistication (Feng and Seasholes, 2005; Seru, Shumway and Stoffman, 2010); once we split realized outcomes by valence, gains and losses move the bias in opposite directions, and losses amplify it rather than tame it. The finding also runs against a belief-pessimism reading, under which painful losses would make investors quicker to realize losses and so weaken the bias; we find the reverse, which is difficult to reconcile with a simple pessimistic-belief account and consistent with memory.

Second, we give the disposition effect a memory microfoundation. Adapting the temporal-context model of Howard and Kahana (2002) and the memory-and-decisions framework of Malmendier and Wachter (2021), we show that one cued-recall channel reproduces the baseline effect and both experience patterns.

Third, we relate the model to other documented patterns in selling behavior. The same channel delivers the V-shaped selling propensity of Ben-David and Hirshleifer (2012): large returns of either sign are the most retrievable, so the most extreme positions are the ones most likely to be sold. The rank effect of Hartzmark (2015) points the same way: the best- and worst-performing positions in a portfolio are the ones most likely to be sold, and they are also the most retrievable. Because diversification reduces the right tail of the return distribution, the model predicts a weaker disposition effect for mutual funds; Chang, Solomon and Westerfield (2016a) show that the same investors who exhibit a positive disposition effect in individual stocks exhibit a negative one in mutual funds. The channel also fits direct evidence on investor memory: recalled returns move with market conditions (Jiang et al., 2024), and the causal result of Frydman and Wang (2020) shows that making the purchase

price more prominent on a brokerage screen raises the disposition effect.

The closest prior work is Ben-David and Hirshleifer (2012). They show that the probability of selling a stock is V-shaped in its return since purchase, rising with the size of gains and, at short holding periods, with the size of losses, with the gain arm steeper throughout and little evidence of a jump at zero, and that the V flattens as a position ages; they read the pattern as consistent with trading on beliefs revised by large price moves rather than with a preference for realizing gains. Our mechanism reproduces the V without a belief step and locates the disposition effect in what comes to mind at the moment of sale. The dependence on holding period may reflect recency in recall; the model presented here abstracts from recency and does not derive this pattern. We return to it in Section 4 and leave its formal treatment to future work.

The paper proceeds as follows. Section 2 describes the brokerage data, the 2013–2016 market cycle, and how we measure winning and losing experiences. Section 3 presents the core result, the opposite-signed response of the disposition effect to gains and losses, and shows it survives alternative thresholds and a ratio specification. Section 4 develops the memory-based recall model, maps it to the evidence, and draws out its further implications—the V-shaped selling schedule, the rank effect, evidence on what investors have in mind when they sell, including holding period, and mutual funds. Section 5 examines alternative mechanisms: a mechanical artifact of portfolio composition, unobserved skill and portfolio-level effects, regression to the mean, return extrapolation, portfolio rebalancing, and overconfident overtrading. Section 6 tests one prediction of the model: the experience effect is stronger for younger investors, whose store of competing memories is smaller. Section 7 concludes.

2 Data and Methodology

2.1 Retail Investment Data

We draw on transaction-level records from a major Chinese securities brokerage spanning early July 2013 to late February 2016. The unit of observation is a single trade: over 40 million transactions by 189,530 individual retail investors trading roughly 4,000 stocks. Ta-

ble 1 reports investor descriptive statistics. Each record carries the number of shares bought and sold, the purchase and sale prices, and the prevailing market price. Linking records to account IDs adds investor demographics—age, gender, and education.¹ The brokerage also records each investor’s mandatory risk-tolerance assessment, completed at account opening.

2.2 Bull and Bear Markets

Figure 1 places our sample period in market context, plotting the Chinese stock market over the span of our data. Unlike datasets confined to bull markets, our window covers bull, bear, and low-volatility regimes. Investors in our sample therefore face widely varying odds of realizing gains, losses, or a mix of the two—the cross-investor variation our design needs.²

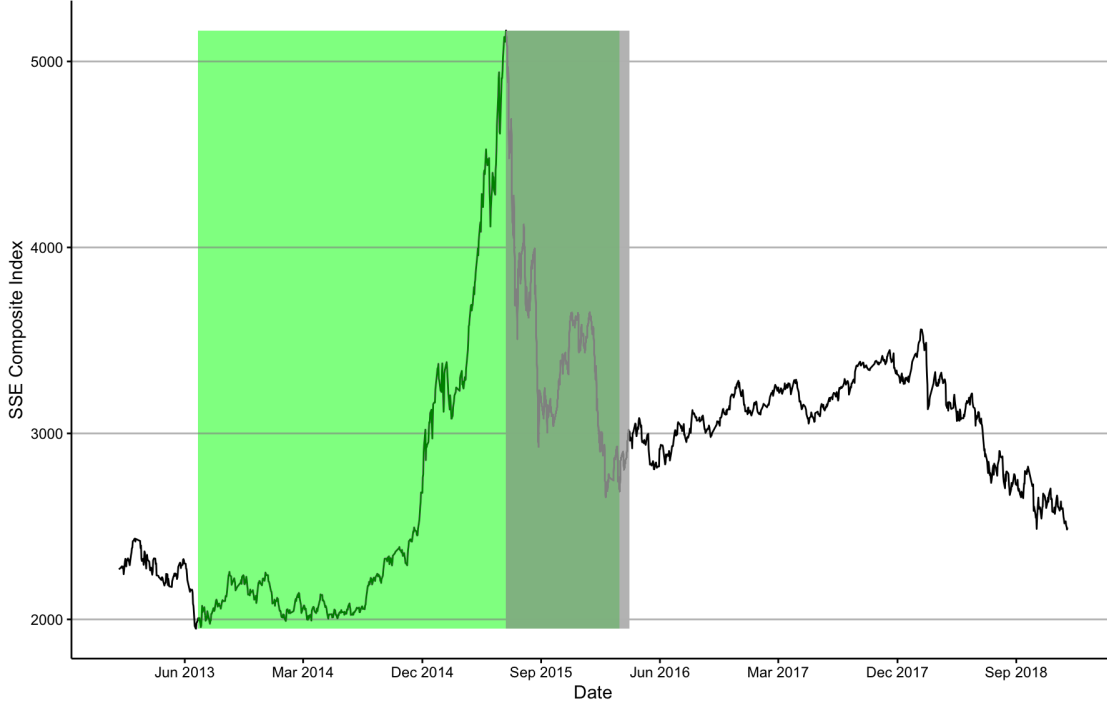
2.3 Constructing the Key Variables

We build our panel at the investor–stock–day level: on each date an investor sells at least one security, we record the position sold along with every other position the investor holds that day. The regression sample contains 43,649,867 investor–stock–day observations. Dropping accounts opened before our first observation date, July 1, 2013, restricts the sample to accounts whose complete trading history we observe; average account tenure is 340 days, measured from account opening through the last observable date, February 29, 2016. We describe each step below.

¹Under China’s real-name registration system, each securities account is tied to a single national identity card, and an investor cannot hold more than one account at the same brokerage. The account we observe therefore reflects the investor’s complete holdings at this broker—the relevant unit for measuring the disposition effect and portfolio composition. An investor may in principle maintain a separate account at another brokerage. Unobserved outside positions and trades introduce measurement error both in account-wide portfolio variables and in our experience counts, which capture within-brokerage rather than total personal trading history; if this error is approximately classical, it attenuates the estimated experience effects toward zero.

²Many of the most relevant studies draw on predominantly bull-market periods—for example, Feng and Seasholes (2005) in China and Barber and Odean (2000a) in the United States both cover sustained upswings—whereas our 2013–2016 window spans a full boom and bust, with the index rising from about 2,000 to 5,000 before falling back below 3,000.

Figure 1: Chinese Bull and Bear Market, 2013–2016



Notes. This figure illustrates the performance of the SSE Composite Index over a specified period. The green area represents the period covered by our data, while the gray area marks the period identified as a market crash. The SSE Composite Index, which includes all A-shares and B-shares listed on the Shanghai Stock Exchange, reflects the overall performance of the Chinese stock market. The horizontal axis is the calendar date (July 2013 to February 2016) and the vertical axis is the daily closing index level. Source: Shanghai Stock Exchange.

2.3.1 Portfolio Positions and Investment Gains and Losses

Positions. We sort the data by account ID, then stock ticker, then date. Following Odean (1998) and subsequent disposition-effect studies (Feng and Seasholes, 2005; Seru, Shumway and Stoffman, 2010), we define a position as beginning when an investor first purchases a given stock and ending at the first sale of that stock, and we treat any shares remaining after a partial sale as the start of a new position.³ An investor can therefore reinvest in the same stock many times within a position; when this happens, we set the position’s cost basis to the volume-weighted average of all purchase prices, so the realized return at sale is measured against that average. Table 1 reports descriptive statistics for the holdings and trading experiences. The average investor holds 7.6 stocks, and portfolio values are strongly

³We find similar results if partial sales are excluded from our sample.

Table 1: Descriptive Statistics of Investors, Holdings, and Trading Experiences

	Percentiles				
	Mean	SD	25th	50th	75th
<i>Investors and Accounts</i>					
Number of investors	189,530	-	-	-	-
Account tenure (days)	339.90	162.00	266	308	362
Number of account-stock-days	43,649,867	-	-	-	-
Percentage of male investors (%)	55%	-	-	-	-
Average investor age (years)	37.88	11.06	29	36	45
Risk-taking level (1-5)	3.05	0.74	3	3	3
Education level (1-7)	4.11	1.28	3	4	5
<i>Holdings</i>					
Number of stock holdings	7.59	9.24	2.27	4.83	9.52
Portfolio value (RMB)	358,875	3,357,000	19,467	63,472	206,929
<i>Trading Experiences</i>					
Losing experiences	12.40	28.30	0	4	13
Losing experiences (5%)	6.45	13.39	0	2	8
Losing experiences (30%)	0.68	1.92	0	0	1
Winning experiences	14.79	35.95	1	5	16
Winning experiences (5%)	5.66	12.90	0	2	7
Winning experiences (30%)	0.28	1.49	0	0	0

Notes. Risk-taking level 1 represents the most risk-averse and 5 the most risk-seeking investors. Education level 1 represents the highest level of education (PhD degree) and 7 represents the lowest level of education (elementary school level). The number of stocks held is calculated at the trading day level, reflecting the average number of stocks an investor has during selling transactions. Portfolio value is calculated using the same approach. Losing and winning experiences represent the number of respective experiences an investor had. RMB stands for RenMinBi, the currency of mainland People’s Republic of China (PRC). Trading experiences are measured by the number of realized returns, where each negative realized return is counted as one losing experience. The 5% and 30% experience measures count only prior trades whose realized returns (in absolute value) exceed 5% and 30%, respectively, relative to the investor’s average purchase price. The sample period is July 2013 to February 2016, covering 189,530 investor accounts; the number of account-stock-days reported here is the same quantity used as the number of observations in the regression analysis. Demographic variables (gender and education) are missing for some accounts ($N = 128,563$ and $122,452$, respectively), so heterogeneity analyses involving these characteristics run on the corresponding sub-samples. The trading-experience and portfolio-value distributions are heavily right-skewed; standard deviations exceed means due to a small hyperactive tail (e.g., the maximum Losing Experience is 2,563).

right-skewed.

Gains and Losses. Following Odean (1998), we evaluate a portfolio’s gains and losses only on dates when the investor sells some security, because on no-sale days we cannot tell deliberate holding from inattention (Chang, Solomon and Westerfield, 2016b). For each

investor–stock–day we compare the relevant price to the cost basis and classify the position as a gain ($\text{Gain} = 1$) when the return is positive and a loss ($\text{Gain} = 0$) otherwise. For the stock actually sold, we use the realized return computed at the sell order price; for every other stock held that day, we use the unrealized (‘paper’) return computed at the day’s opening price.^{4,5}

A losing experience is a completed sale that realizes a return below a threshold relative to the investor’s average purchase price, and a winning experience one above it; an investor’s losing and winning experiences are the running counts of each. We vary the threshold from 5% to 30%; results are robust across these cutoffs. Table 1 shows that investors record 12 losing and 15 winning experiences on average, the gap reflecting a sample that spans somewhat more bullish than bearish months.

Figure 2 plots the account-level distribution of winning and losing experiences. At exactly one experience, accounts with a single loss far outnumber those with a single win—many investors exit after a first loss, leaving accounts stranded with one losing experience and nothing more. Above that point, the counts of accounts track each other closely across comparable levels of winning and losing experience.

2.3.2 Measuring Trading Experience

Account records avoid survey self-report error but do not reveal how investors perceive their experience. The trade-count measure treats experience as trading activity; the separate winning- and losing-experience measures treat it as realized outcomes.

We begin from the trade-count measure of Dhar and Zhu (2006), which proxies experience by accumulated trading activity.⁶ Dhar and Zhu (2006) reads this measure as learning: List (2003) find experimentally that repetition breeds familiarity, so traders who act more often grow more sophisticated and less prone to bias, and Dhar and Zhu (2006) carries the same logic to investors. Both report that a higher cumulative trade count predicts weaker bias.

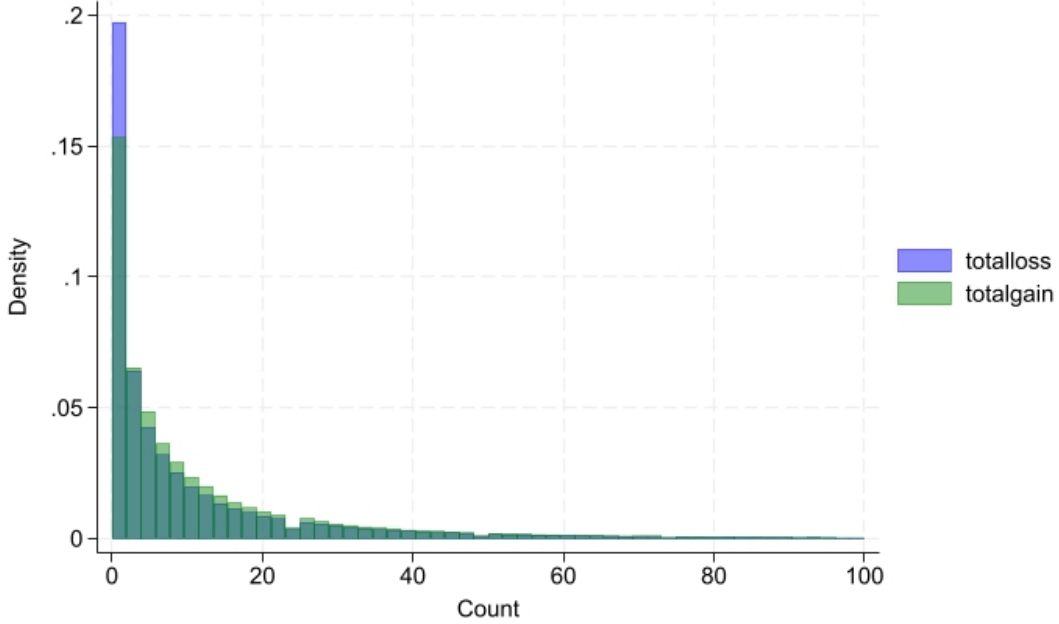
We use this measure as a benchmark. Neither List (2003) nor Dhar and Zhu (2006) can

⁴We use the opening price because it is predetermined relative to the day’s sale decisions; the closing price is not.

⁵Details of our per-transaction return calculations are provided in Appendix A1.

⁶The construction of trading episodes and the associated experience measures is described in detail in Appendix A2.

Figure 2: Distribution of Trading Experience



Notes. This figure shows the distribution of trading experiences across accounts in our sample. The green bars represent each account’s total number of winning experiences (“totalgain”), while the purple bars reflect the total number of losing experiences (“totalloss”). These counts are measured cumulatively as of the end of the observation period (i.e., at time T). Although trading experience is modeled as a time-varying variable in our empirical analysis, the figure presents a static snapshot of each account’s experience as of the final period. The horizontal axis is the number of trading experiences and the vertical axis is the number of accounts.

separate a winning trade from a losing one—a distinction the sports-card setting of [List \(2003\)](#) cannot support—so each count pools outcomes of opposite valence. We show instead that past wins and past losses act on the current sell decision through distinct, opposing channels. We first replicate the pooled trade-count result, then split experience by outcome to show how past realized outcomes bear on the current decision.

Distinguishing Win and Loss Experiences. We start from the pooled count above, then separate experiences by valence, defining:

$$\text{Loss Experience}_{i,t} = \sum_{T=0}^{t-1} \mathbf{1}\{\text{return}(s, j)_T < -X\}$$

$$\text{Win Experience}_{i,t} = \sum_{T=0}^{t-1} \mathbf{1}\{\text{return}(s, j)_T > X\}$$

where $\mathbf{1}\{\cdot\}$ denotes a loss (win) experienced by investor i at time T for stock j , and X is the threshold above which a realized gain or loss counts as an experience. We vary X for robustness and adopt $X = 5\%$ and $X = 30\%$ in our primary analysis.

Because more active investors mechanically accumulate more losses in absolute terms, we also scale by total activity, using the share of losing experiences among all experiences:

$$\text{Loss Experience Share}_{i,t} = \frac{\sum_{T=0}^{t-1} \mathbf{1}\{\text{return}(s, j)_T < -X\}}{\text{Number of prior sales}_{i,t}} \quad (1)$$

A simple example: when $X = 0$, if an investor has 15 selling transactions before making the current decision and 5 of them have positive realized returns, then $\text{Loss Experience Share}_{i,t} = 2/3$.

2.4 Measuring the Disposition Effect

We measure the disposition effect two ways: an investor-level proportion and a position-level regression coefficient. Following [Dhar and Zhu \(2006\)](#), we start with the simplest investor-level measure. Following [Odean \(1998\)](#), we define the Proportion of Gains Realized (PGR) and the Proportion of Losses Realized (PLR) as

$$\begin{aligned} \text{PGR} &= \frac{\text{Realized Gains}}{\text{Realized Gains} + \text{Paper Gains}} \\ \text{PLR} &= \frac{\text{Realized Losses}}{\text{Realized Losses} + \text{Paper Losses}} \\ \text{DE} &= \text{PGR} - \text{PLR} \end{aligned} \quad (2)$$

Realized Gain (Loss) counts the stocks sold in a portfolio at a positive (negative) realized return; Paper Gain (Loss) counts the stocks held at a positive (negative) unrealized return at the time of calculation. Equation 2 is the standard measure of the disposition effect's magnitude: $\text{DE} > 0$ signals the disposition effect, a higher propensity to realize gains than losses.

We next quantify that magnitude with the regression approach of [Odean \(1998\)](#), regressing a stock-position sell indicator (1 = Sell; 0 = Hold) on a *Gain* indicator (1 = the stock trades at an unrealized gain; 0 otherwise) among other controls. Its coefficient measures how much more likely investors are to sell a winner.

3 Experience-Driven Disposition Effect

This section takes the memory model’s three qualitative predictions to the data: a disposition effect already present at zero measured large-outcome experience, its attenuation as realized gains accumulate, and its amplification as realized losses accumulate. Section 4 derives all three from a single retrieval channel. The first is plain in the raw data: across all paper returns recorded on sale dates, investors realize 12.4% of gains but only 8.3% of losses (9.7% overall), in line with [Odean \(1998\)](#) and subsequent studies.

We estimate a linear probability model (LPM) with high-dimensional fixed effects, following [An et al. \(2024\)](#). Investor, stock, and time fixed effects absorb unobserved heterogeneity across investors, persistent differences across stocks, and common time variation such as the boom–bust cycle. We cluster standard errors three ways—by investor, day, and stock ([Cameron, Gelbach and Miller, 2011](#))—letting the sale residuals correlate arbitrarily within an investor, within a day, and within a stock. Because experience varies only at the investor level, investor clustering binds for the experience interactions: collapsing to two-way (investor and day) or one-way (investor) clustering leaves their standard errors essentially unchanged (untabulated).

3.1 A Benchmark: Trading Experience and the Disposition Effect

We begin by confirming that our data reproduce the benchmark pattern in [Feng and Seasholes \(2005\)](#), estimating a regression close to theirs⁷:

$$\text{Sale}_{i,j,t} = \alpha + \beta_1 \cdot \text{Gain}_{i,j,t} + \beta_2 \cdot \text{Experience}_{i,t} + \beta_3 \cdot \text{Gain}_{i,j,t} \times \text{Experience}_{i,t} + \epsilon_{i,j,t} \quad (3)$$

⁷While they employ a hazard model, we use linear probability model (LPM) in our analysis to demonstrate that we observe the same pattern in our results.

Table 2: Trading Experience and Disposition Effect

Dependent Variable:	Sale $\times$ 100	
	(1)	(2)
Gain	2.5037*** (4.74)	2.7506*** (4.77)
Total Experience	0.0025* (1.83)	0.0029** (2.06)
Gain $\times$ Total Experience		-0.0020** (-2.58)
Constant	8.5853*** (34.88)	8.5345*** (33.44)
Observations	43,649,867	43,649,867
R-squared	0.139	0.139

Notes. The table reports results for the linear probability model shown in equation 3 on our sample. *Total Experience* is the cumulative number of prior trades (wins or losses) executed by the investor before the current sell decision. Column (1) omits the *Gain $\times$ Total Experience* interaction; Column (2) adds it, capturing how the disposition effect varies with overall trading experience. The outcome dummy is multiplied by 100 to improve the readability of the coefficients. Both columns include investor, stock, and time fixed effects, and standard errors are clustered by investor, time, and stock. The sample period is July 2013 to February 2016. Numbers in parentheses are t-statistics. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

$\text{Sale}_{i,j,t}$ equals 1 if stock j in account i is sold, in whole or in part, at time t , and 0 otherwise, so the fitted value is investor i 's willingness to sell asset j at time t . $\text{Gain}_{i,j,t}$ equals 1 if the market price of asset j exceeds its average purchase price at time t (a positive unrealized return), and 0 otherwise, so β_1 measures the propensity to sell a winner. We measure experience as

$$\text{Experience}_{i,t} = \text{Number of sales taken by investor } i \text{ up until date } t.$$

The coefficient of interest is β_3 on the interaction $\text{Gain}_{i,j,t} \times \text{Experience}_{i,t}$. Because *Gain* is binary, experience shifts the sale probability by β_2 for a losing position and by $\beta_2 + \beta_3$ for a winning one, so β_3 measures how the disposition effect varies with experience: a positive coefficient indicates amplification and a negative coefficient attenuation.

We report the estimates in Table 2. Column (2) shows that the coefficient on *Gain* is positive and significant, confirming the disposition effect among retail investors, while the interaction between *Gain* and *Total Experience* is negative and significant, so the disposition effect attenuates as investors accumulate experience.

The disposition effect thus weakens as investors accumulate trading experience, consistent

Table 3: The Experience-Driven Disposition Effect

Dependent Variable:	Sale $\times$ 100				
$ X > 5\%$	(1)	(2)	(3)	(4)	(5)
	Main Effects		Interactions		Full Spec.
Gain	2.4936*** (4.74)	2.4963*** (4.74)	2.4765*** (4.34)	2.6628*** (4.79)	2.4781*** (4.35)
Loss Experience	0.0163*** (3.17)		0.0162*** (3.12)		0.0109 (1.13)
Win Experience		0.0087 (1.31)		0.0103 (1.50)	0.0034 (0.36)
Gain $\times$ Loss Experience			0.0007 (0.30)		0.0212*** (3.28)
Gain $\times$ Win Experience				-0.0068*** (-2.66)	-0.0198*** (-3.30)
Constant	8.4347*** (37.46)	8.6229*** (35.03)	8.4384*** (36.15)	8.5822*** (33.86)	8.4974*** (36.63)
Observations	43,649,867	43,649,867	43,649,867	43,649,867	43,649,867
R-squared	0.138	0.138	0.138	0.138	0.138

Notes. Estimates from the linear probability model in equation 3; the dependent variable is an indicator for sale, multiplied by 100. *Loss Experience* and *Win Experience* count the investor’s prior trades that realized a loss or a gain exceeding 5% relative to the average purchase price. Columns (1)–(2) include only the main effects for *Loss Experience* and *Win Experience*; columns (3)–(4) add interactions between *Gain* and each count; column (5) reports the full specification with both counts and both interactions. *Gain* equals 1 if the position has an unrealized gain at the decision time. All specifications include investor, stock, and time fixed effects, and standard errors are clustered by investor, time, and stock. The sample period is July 2013 to February 2016. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

with the pattern documented by [Feng and Seasholes \(2005\)](#).⁸

3.2 Trading Experience with Both Wins and Losses

Investors with the same number of prior trades can have different histories of gains and losses. We therefore measure winning and losing experience separately. We define a *losing* (*winning*) *experience* as a sale that realizes a return below -5% (above $+5\%$) relative to the investor’s average purchase price. Our results are robust to alternative cutoffs⁹.

We split prior experience by valence in Table 3. The baseline disposition effect holds across every column: the coefficient on *Gain* is positive and significant at the 1% level, so investors sell winners more readily than losers. Winning and losing experience move that

⁸[Feng and Seasholes \(2005\)](#) additionally allow this effect to be non-linear, using dummies for bins of accumulated trades (fewer than five prior trades, fewer than ten, and so on). Because we separate winning from losing experiences and require each realization to clear a return threshold, our experience counts are far lower than theirs, leaving a binned specification little power to detect non-linearity; we therefore do not pursue it.

⁹We test the robustness of our results using alternative thresholds of 5%, 10%, 15%, 20%, 25%, and 30%. For clarity of presentation, we primarily report results based on the 5% threshold, while also reporting the 30% threshold to capture only the largest outcomes.

tendency in opposite directions.

Column (5) reports the full specification. The interaction of *Gain* with *Loss Experience* is positive and significant at the 1% level (0.0212, $t = 3.28$): the disposition effect strengthens as losing experiences accumulate. A two-standard-deviation increase in *Loss Experience* (26.78) raises the propensity to sell a winner rather than a loser by 0.57 percentage points—about 23% of the 2.48-percentage-point baseline *Gain* effect.

The interaction of *Gain* with *Win Experience* is negative and significant at the 1% level (-0.0198 , $t = -3.30$), indicating that the disposition effect weakens as winning experiences accumulate. A two-standard-deviation increase in *Win Experience* (25.80) cuts the relative propensity to realize winners by 0.51 percentage points, roughly 21% of the baseline. In per-trade terms, each additional large realized loss widens the winner–loser sale gap by about 0.02 percentage points and each additional large gain narrows it by the same amount, just under 1% of the baseline gap apiece.

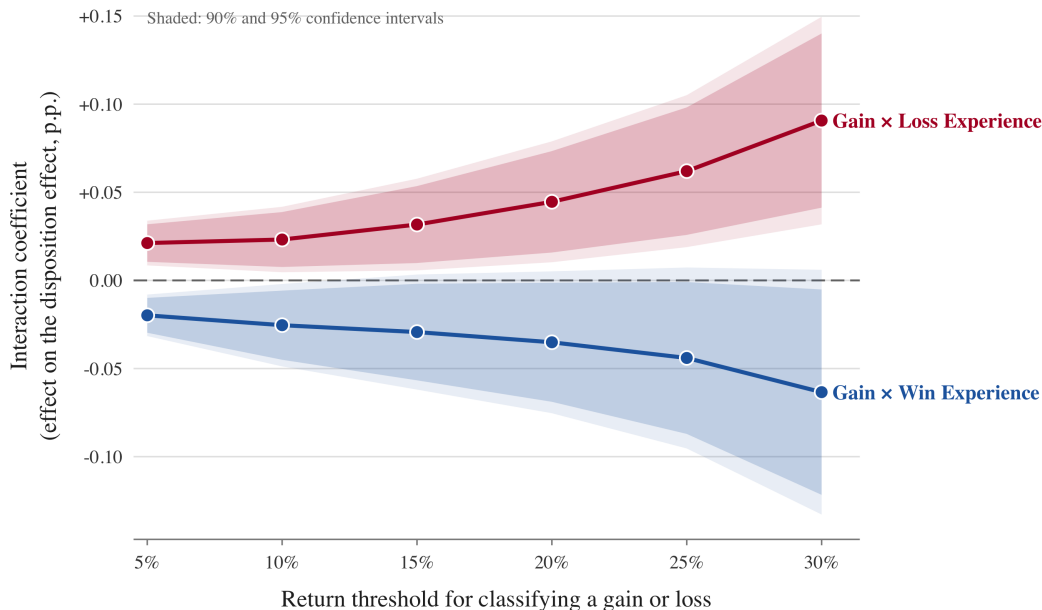
Two features of the column progression matter. First, the attenuation is already visible when the win interaction enters alone (column (4): -0.0068 , $t = -2.66$), while the loss interaction is small and insignificant on its own (column (3): 0.0007 , $t = 0.30$); entering both jointly sharpens the valence-specific asymmetry. Second, either channel is an order of magnitude larger than the attenuation the pooled count delivered in Table 2 (-0.0020): pooling nets two opposing forces against each other and understates both.

3.3 Varying the Threshold for Wins and Losses

We count only outcomes beyond a 5% threshold as experiences; the asymmetry does not hinge on that choice. Figure 3 plots the $\text{Gain} \times \text{Loss Experience}$ and $\text{Gain} \times \text{Win Experience}$ coefficients as the classification threshold varies from 5% to 30%. Both interactions keep their signs across the full range, the win side negative and the loss side positive. Their per-event magnitudes grow as the threshold rises, though part of that growth is mechanical—a higher cutoff counts rarer events, so each counted experience carries more weight—and we therefore read the figure as evidence of sign stability rather than as a salience gradient. Table 4 reports the full specification at the extreme 30% cutoff.

We label the two patterns for reference. The *fear effect* is the loss-side amplification:

Figure 3: Experience–Gain Interactions Across Return Thresholds



Notes. The figure plots the two interaction coefficients—*Gain × Loss Experience* (red, fear) and *Gain × Win Experience* (blue, greed)—estimated jointly from the full specification in Column (5) of Table 3. The level effects are not plotted. The horizontal axis is the threshold $|X|$ above which a realized gain or loss counts as an experience, varied from 5% to 30%; a higher threshold counts only more extreme realizations. Shaded regions are 90% (darker) and 95% (lighter) confidence intervals, computed from standard errors clustered by investor, time, and stock.

investors who have realized large losses sell winners relative to losers more readily. The *greed effect* is the win-side attenuation: investors with more large realized gains exhibit a smaller gap between the probabilities of selling winners and losers. The labels are descriptive, not explanations. Section 4 derives both patterns from cued recall, and Section 5 examines explanations based on extrapolative optimism and overconfidence.

3.4 Trading Experience as a Ratio

We next measure experience as a share rather than a count. Counts conflate valence with volume: more active investors accumulate more winning and more losing experiences alike, so each count partly tracks overall activity rather than the balance of past outcomes. The share of prior trades that ended in a large loss (gain) holds activity fixed and asks only about the mix (Equation 1).

Table 4: Experience-Driven Disposition Effect (30% Threshold)

Dependent Variable:	Sale $\times$ 100	
$ X > 30\%$	(1)	(2)
Gain	2.5063*** (4.74)	2.4275*** (4.42)
Loss Experience	0.0912* (1.90)	0.0657 (1.19)
Win Experience	-0.0035 (-0.04)	0.0147 (0.14)
Gain $\times$ Loss Experience		0.0907*** (3.02)
Gain $\times$ Win Experience		-0.0634* (-1.79)
Constant	8.7203*** (44.35)	8.7541*** (43.88)
Observations	43,649,867	43,649,867
R-squared	0.139	0.139

Notes. Linear probability model where the dependent variable equals 1 if the position is sold (multiplied by 100). *Loss Experience* and *Win Experience* count prior trades with realized returns below -30% or above $+30\%$, respectively, relative to the investor’s average purchase price. Coefficients are reported with t -statistics in parentheses. All specifications include investor, stock, and time fixed effects, and standard errors are clustered by investor, time, and stock. The sample period is July 2013 to February 2016. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 5 re-estimates the same linear probability model, replacing the experience counts with the losing- and winning-experience shares (Equation 1). The winning-versus-losing contrast sharpens. In Column (2), the *Gain $\times$ Loss Experience Share* interaction enters at 32.37 ($t = 11.63$), significant at the 1% level: moving from a loss-free history to an all-loss one—the extremes of the share—widens the gain–loss gap in selling by roughly 32 percentage points, against a baseline *Gain* coefficient of 1.80. The winning share pulls the other way. In Column (4), the *Gain $\times$ Win Experience Share* interaction is -5.93 ($t = -3.72$), significant at the 1% level, so a loss-heavy past strengthens the disposition effect while a win-heavy past dampens it¹⁰. Column (5) enters both shares jointly and reproduces the pattern, with the two interactions at 32.49 ($t = 11.66$) and -6.80 ($t = -4.34$). The asymmetry survives the move from counts to shares, so it tracks the relative mix of winning and losing experiences, not the number of trades behind them.

¹⁰At low thresholds nearly every realized trade clears the cutoff and so counts as a win or loss, so the two shares sum to approximately one, $s^{\text{win}} + s^{\text{loss}} \approx 1$ (exactly one when no sub-threshold trades remain). The shares—and their *Gain* interactions—are then mutually near-collinear (and each near-collinear with the constant), leaving the joint specification in Column (5) ill-conditioned. We therefore measure experience shares at the 30% threshold, where a substantial fraction of trades fall below the cutoff: this restores independent variation in the two shares and lets both enter jointly without inflating their standard errors. The contrast is not specific to this threshold—the count-based specification of Table 3, in which the winning- and losing-experience counts are not collinear, identifies the two channels separately at the 5% threshold.

Table 5: The Experience-Driven Disposition Effect: Share of Losing vs. Winning Experiences

Dependent Variable:	Sale $\times$ 100				
$ X > 30\%$	(1)	(2)	(3)	(4)	(5)
Gain	2.4502*** (4.64)	1.8020*** (3.36)	2.4504*** (4.64)	2.5127*** (4.71)	1.8712*** (3.45)
Loss Experience Share	3.9275*** (5.93)	-3.0334*** (-2.89)			-2.9903*** (-2.85)
Gain $\times$ Loss Experience Share		32.3716*** (11.63)			32.4871*** (11.66)
Win Experience Share			2.7827*** (3.45)	6.0898*** (4.67)	6.8336*** (5.29)
Gain $\times$ Win Experience Share				-5.9286*** (-3.72)	-6.7951*** (-4.34)
Constant	8.4407*** (44.39)	8.6346*** (45.11)	8.4973*** (45.73)	8.4649*** (44.90)	8.5648*** (44.05)
Observations	42,969,625	42,969,625	42,969,625	42,969,625	42,969,625
R-squared	0.127	0.127	0.127	0.127	0.127

Notes. This table reports estimates from the linear probability model described in equation 3, using an alternative measure of trading experience based on the share of large losing or winning trades. The dependent variable is a dummy indicating whether a stock is sold, multiplied by 100. Columns (1)–(2) use *Loss Experience Share* – the proportion of prior trades that ended in a large loss; Columns (3)–(4) use *Win Experience Share*; Column (5) includes both shares jointly. The interaction terms *Gain* $\times$ *Loss Experience Share* and *Gain* $\times$ *Win Experience Share* capture how the disposition effect varies with the composition of past experiences. Experience shares are measured at the 30% threshold: at lower thresholds nearly all trades clear the cutoff, so $s^{\text{win}} + s^{\text{loss}} \approx 1$ and the two shares cannot be separately identified in Column (5), whereas the count-based results of Table 3 establish the same winning-versus-losing contrast at the 5% threshold. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock. The sample period is July 2013 to February 2016. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

4 A Memory-Based Model

In this section, we develop a model of cued recall in the context of stock market decision-making, drawing on the frameworks of Howard and Kahana (2002) and Kahana, Paron and Wachter (2025). We then extend the model to align with and explain the key empirical patterns documented in our analysis.

The standard account of the disposition effect relies on prospect theory. The literature has pointed out numerous problems with this account, including the requirement of specifying a reference point,¹¹ and explaining why investors would have a utility function over returns, and in particular, returns that are converted into cash (sometimes called “realization utility”).

¹¹Prospect theory has its origin in explaining subject’s responses to lottery-type questions in the laboratory, where it is necessary methodologically to define a reference point (Luce, 2000), utility over gains and losses. At some point, this methodological point became a theory about the world, though on what basis was not made clear.

The more recent evidence on the diminishment of the disposition effect when accounting for experience increases these problems. While it is plausible that investors may exhibit more rational behavior after learning from their mistakes, the premise of prospect theory is that the behavior is rational because it originates in the utility function. Learning about having a different utility function is at present outside of economics.

Recent models have gotten around this difficulty by asserting that utility parameters change, and through experience (e.g., [Gennaioli and Shleifer \(2010\)](#), [Barberis \(2012\)](#), [Barberis and Xiong \(2012\)](#)). This account does not explain why the parameters would change in a particular way (nor, of course, does it explain why “realization utility” over losses is how investors think about their portfolios).

The evidence in the present paper strains this approach even further, for it fails to explain why winning experiences versus losing experiences would affect the utility function differently.

Here we present a single mechanism, based on representativeness combined with skewness in underlying stock returns, that accounts for the baseline disposition effect, its diminution under the experience of gains, and its increase under the experience of large losses. Our focus is, however, on explaining the new evidence in this paper, acknowledging that there might be other explanations for the disposition effect itself, particularly when skewness is taken into account.

The decision of which stock to sell resembles a cued recall task, with “stocks I own” forming the cue. We use the version of the [Howard and Kahana \(2002\)](#) temporal context model in [Kahana, Paron and Wachter \(2025\)](#), applied to cued recall rather than associative recognition.

Agents view features f_i : basis vectors in a high-dimensional vector space. These features become embedded in memory together with a temporal context, an internal mental state that is persistent and evolves through time. Context, through the memory matrix, governs what comes to mind. Specifically, given an initial condition M_0 , memory is comprised of

outer products of features and contexts $\{f_i, x_i\}_{i=1}^n$:

$$M_n \equiv M_{n-1} + x_n f_n^\top \quad (4)$$

$$= M_0 + \sum_{i=1}^n x_i f_i^\top \quad (5)$$

In what follows, we will be interested in a process known as cued recall, which has long been studied in the literature on human memory. Context is retrieved in response to a features cue:

$$\mathbf{x}_{\text{CUE}} \propto M_n f_{\text{CUE}}, \quad (6)$$

where context is scaled to have a norm of 1. Substituting (6) into (5) yields:

$$\mathbf{x}_{\text{CUE}} \propto M_0 f_{\text{CUE}} + \sum_{i=1}^n x_i (f_i^\top f_{\text{CUE}}). \quad (7)$$

Equation 7 shows that features evoke the past contexts under which they are experienced. A context x appears in the sum in (7) if the corresponding f_i equals the current feature; otherwise it does not. Thus context is a weighted average of past contexts under which that feature was experienced. In what follows, we assume M_0 is the zero matrix and we suppress the n subscript on M . Context x_i is endogenous and persistent, and will be described in more detail in what follows.

We use well-established principles of association to determine what pops to mind. As mentioned above, we think of the decision as a cued recall task. In a cued recall task, the first step is to retrieve the context associated with the physical features that form the cue as in (6). Unlike in a recognition task, where it is sufficient to judge the similarity of the cued context with a target context,¹² here features are retrieved based on the cue as in Howard and Kahana (2002):

$$\mathbf{f}_{\text{CUE}} \propto M^\top \mathbf{x}_{\text{CUE}}. \quad (8)$$

These are features that are experienced under contexts similar to $\mathbf{x}_{\text{CUE}}$. There will certainly be nonzero weight on the element corresponding to f_{CUE} , namely $f_{\text{CUE}} \cdot \mathbf{f}_{\text{CUE}} \neq 0$. However,

¹²See Kahana, Paron and Wachter (2025) for a discussion of recall versus recognition

any feature experienced at the same time as, or close in time to, f_{CUE} will also be recalled. This is the core mechanism of the temporal context model of Howard and Kahana (2002).

Each candidate feature f_i likewise retrieves a context

$$\mathbf{x}_i \propto M f_i. \quad (9)$$

Like $\mathbf{x}_{\text{CUE}}$, each retrieved context is scaled to have a norm of one; throughout, the norm is the L^2 norm. This per-candidate normalization is the source of the $1/\sqrt{n_i}$ factors in Theorem 2 below, where n_i denotes the number of observations of f_i .

The vector of retrieved features, $\mathbf{f}_{\text{CUE}}$, has nonzero weight on a number of elements. For one of them to rise above the others, it first needs to have a nonzero weight in $\mathbf{f}_{\text{CUE}}$, and relatively speaking, a similar context to $\mathbf{x}_{\text{CUE}}$. Because recall is competitive, if the context is sufficiently different from $\mathbf{x}_{\text{CUE}}$, recall of that feature will be suppressed. The following model captures these properties. The competition runs over feature types rather than individual episodes: multiple episodes may share the same feature, and each of the N distinct feature types stored in memory enters the competition once. For $i = 1, \dots, N$,

$$\mathbb{P}(\text{recall of } f_i) \propto (\mathbf{x}_i \cdot \mathbf{x}_{\text{CUE}}) \|\mathbf{f}_i \cdot \mathbf{f}_{\text{CUE}}\|_0.$$

Equivalently,

$$\mathbb{P}(\text{recall of } f_i) = \frac{(\mathbf{x}_i \cdot \mathbf{x}_{\text{CUE}}) \|\mathbf{f}_i \cdot \mathbf{f}_{\text{CUE}}\|_0}{\sum_{j=1}^N (\mathbf{x}_j \cdot \mathbf{x}_{\text{CUE}}) \|\mathbf{f}_j \cdot \mathbf{f}_{\text{CUE}}\|_0}. \quad (10)$$

Note that $\|\mathbf{f}_i \cdot \mathbf{f}_{\text{CUE}}\|_0$ is the zero norm; it is either equal to zero or one, zero if f_i is not recalled in (8). This specification is a simplified version of the retrieval process in Lohnas, Polyn and Kahana (2015).

We use the model in Kahana, Paron and Wachter (2025) for context evolution:

$$x_{i+1} \begin{cases} = & x_i \text{ if no distractor} \\ \in & \{x_1, \dots, x_i\}^\perp \text{ otherwise} \end{cases} \quad (11)$$

See Kahana, Paron and Wachter for further details. This formula for context allows for basis vector features that are observed instantaneously to be coded together in context, for

example, a stock that is owned is a feature, as is whether the stock has a gain. This allows for the creation of a composite feature, the sum of two feature vectors observed at (nearly) the same point in time. However, (11) is a highly simplified version of what is believed to be true contextual evolution; therefore, we abstract from well-established principles of recall, such as recency and contiguity.

We now link this memory model to the specific context of the decision to sell gains or losses. The agent’s information set contains owned stocks alongside other commodities. Among these other commodities are stocks the agent has previously sold, which we track separately because selling moves a stock from the owned to the sold category, and it is this reallocation that generates the experience effects below. We therefore distinguish three categories: owned stocks, sold stocks, and other, where sold stocks are a sub-category of the non-owned (“other”) commodities. Before the agent has any selling experience there are no sold stocks, so $n_{\text{SOLD}} = 0$ and the baseline analysis involves only owned stocks and other commodities. Each commodity could have either a large or a small return. Define

$$n_k \equiv \text{number of observations of } k$$

and

$$n_{k,\ell} \equiv \text{number of joint observations of } k \text{ and } \ell$$

for $k, \ell \in \{\text{OWNED}, \text{SOLD}, \text{OTHER}, \text{LARGE}, \text{SMALL}\}$. The cue is “stocks I own.” We use the shorthand notation $\mathbb{P}(\text{LARGE})$ for $\mathbb{P}(\text{recall of } f_{\text{LARGE}})$, and similarly for $\mathbb{P}(\text{SMALL})$.

Assumption 1 *Agents perceive returns as belonging to categories: small and large. Whereas commodities of any type can have small returns, only stocks can have large returns.*

Theorem 1 *Assume that the agent has no winning or losing experiences. The agent competitively recalls either large or small returns with*

$$\begin{aligned} \mathbb{P}(\text{LARGE}) &\propto (\mathbf{x}_{\text{LARGE}} \cdot \mathbf{x}_{\text{OWNED}}) \|\mathbf{f}_{\text{LARGE}} \cdot \mathbf{f}_{\text{OWNED}}\|_0 \\ \mathbb{P}(\text{SMALL}) &\propto (\mathbf{x}_{\text{SMALL}} \cdot \mathbf{x}_{\text{OWNED}}) \|\mathbf{f}_{\text{SMALL}} \cdot \mathbf{f}_{\text{OWNED}}\|_0 \end{aligned}$$

with $\mathbb{P}(\text{LARGE}) + \mathbb{P}(\text{SMALL}) = 1$.

Note that we assume that either LARGE or SMALL is recalled. Of course, the agent will also recall OWNED . As is standard in the memory literature on recall, we assume that the agent suppresses the recall of merely repetitive features.

We apply results in [Kahana, Paron and Wachter \(2025\)](#) to connect the right-hand side quantities to the numbers of observations:

Theorem 2

$$\mathbb{P}(\text{LARGE}) \propto \frac{n_{\text{OWNED,LARGE}}}{\sqrt{n_{\text{OWNED,LARGE}} n_{\text{LARGE}}}} \quad (12)$$

$$\mathbb{P}(\text{SMALL}) \propto \frac{n_{\text{OWNED,SMALL}}}{\sqrt{n_{\text{OWNED,SMALL}} n_{\text{SMALL}}}} \quad (13)$$

Proof. See [Kahana, Paron and Wachter \(2025\)](#). ■

Corollary 1 *If the number of observations of assets other than stocks is sufficiently large, $\mathbb{P}(\text{LARGE}) > \mathbb{P}(\text{SMALL})$.*

Proof. It follows from [Theorem 2](#) that $\mathbb{P}(\text{LARGE}) > \mathbb{P}(\text{SMALL})$ if and only if

$$\frac{n_{\text{OWNED,LARGE}}}{n_{\text{OWNED,SMALL}}} > \sqrt{\frac{n_{\text{LARGE}}}{n_{\text{SMALL}}}} \quad (14)$$

By [Assumption 1](#), $n_{\text{SMALL}} = n_{\text{OWNED,SMALL}} + n_{\text{OTHER,SMALL}} = n_{\text{OWNED,SMALL}} + n_{\text{OTHER}}$, whereas $n_{\text{LARGE}} = n_{\text{OWNED,LARGE}}$. Increasing n_{OTHER} increases n_{SMALL} , and hence decreases the right-hand side of [\(14\)](#), while making no other change. The inequality therefore holds for n_{OTHER} sufficiently large. ■

We now assume that the number of commodities with small price changes is far larger than the number of stocks. This, combined with the fact that stocks exhibit positive skewness (and the memory dynamics) delivers the baseline disposition effect.

Assumption 2 *The agent sells the stock whose returns pop to mind.*

Assumption 3 *Stock returns are positively skewed.*

In our sample, the skewness of paper returns—unrealized gains and losses at the account level—is 23.1, which confirms a highly right-skewed distribution. Consistent with this, [Oh and Wachter \(2022\)](#) provide broader evidence that stock returns generally exhibit positive skewness.

Assumption 4 *The agent creates memories according to (4) and (11).*

Assumption 5 *Cued recall works as follows. The cue retrieves context $\mathbf{x}_{\text{CUE}} \propto M\mathbf{f}_{\text{CUE}}$. This context in turn retrieves the features $\mathbf{f}_{\text{CUE}} \propto M^\top \mathbf{x}_{\text{CUE}}$. Which of these features gets recalled is competitive with probability determined by nonzero weight in $\mathbf{f}_{\text{CUE}}$ and similarity of context associated with the features and the cue as in (10)*

Theorem 3 *The agent is more likely to sell gains than losses.*

Proof. By Theorem 2, the agent is more likely to recall large rather than small returns. It follows from Assumption 3 that these large returns are more likely to be gains. It follows from Assumption 5 that the agent recalls these gains and, finally, from Assumption 2 that the stock with the gain is the stock the agent chooses to sell. ■

The key mechanism is that small returns have contexts associated with “other.” They are not as good a match, in terms of context, to “stock,” and hence less likely to be recalled as part of the cue.

Given this basic structure, the experience effects follow easily. Let $\mathbb{P}_k(\text{LARGE})$ denote the probability of recall of large returns, given k winning experiences, and similarly for $\mathbb{P}_k(\text{SMALL})$.

Theorem 4 *Winning experiences decrease the tendency to sell gains.*

Proof. It follows from Assumption 3 that the large returns are predominantly gains. It suffices to show that winning experiences reduce the ratio $\mathbb{P}_k(\text{LARGE})/\mathbb{P}_k(\text{SMALL})$. After k winning experiences $n_{\text{OWNED,LARGE}}$ is replaced by $n_{\text{OWNED,LARGE}} - k$ whereas $n_{\text{SOLD,LARGE}}$ is replaced with $n_{\text{SOLD,LARGE}} + k$. As a consequence, the total number of owned stocks n_{OWNED} also falls by k , whereas the total number of sold stocks rises by k . It suffices to show that $\mathbb{P}_k(\text{LARGE})/\mathbb{P}_k(\text{SMALL})$ is decreasing in k :

$$\frac{\mathbb{P}_k(\text{LARGE})}{\mathbb{P}_k(\text{SMALL})} = \frac{n_{\text{OWNED,LARGE}} - k}{n_{\text{OWNED,SMALL}}} \sqrt{\frac{n_{\text{SMALL}}}{n_{\text{LARGE}}}}$$

The right-hand side is clearly decreasing in k . ■

Likewise, if the agent sells losses, because these losses tend to be small, this will cement the association between owned stocks and large gains. A losing experience means $n_{\text{OWNED,SMALL}}$ is replaced by $n_{\text{OWNED,SMALL}} - k$. The proof is analogous to Theorem 4.

Theorem 5 *Losing experiences increase the tendency to sell gains.*

The results in the previous sections also indicate that large losses increase the disposition effect more than small losses. While large losses are less likely than large gains, should they occur, they will have the stated effect. We introduce additional feature vectors $f_{\text{LARGE LOSS}}$, $f_{\text{LARGE GAIN}}$, instead of f_{LARGE} . We modify the notation accordingly.

Like large gains, large losses only occur with stocks. We now need to consider three-way comparisons in the theorems above, but given that large gains are more common than large losses, the baseline disposition effect (Theorem 3), the effect of winning experiences (Theorem 4) and the effect of small losing experiences (Theorem 5) still work under this extended model.

As with winning experiences, k large losing experiences implies $n_{\text{OWNED,LARGE LOSS}}$ is replaced by $n_{\text{OWNED,LARGE LOSS}} - k$, whereas $n_{\text{SOLD,LARGE LOSS}}$ is replaced by $n_{\text{SOLD,LARGE LOSS}} + k$. This will decrease the probability $\mathbb{P}(\text{LARGE LOSS})$ by the same reasoning as Theorem 4.

More broadly, because large returns of either sign are the most retrievable—and hence the most likely to be sold—the model implies a V-shaped relation between a position’s realized return and the propensity to sell it, the regularity documented by Ben-David and Hirshleifer (2012).

The model’s results correspond to the three empirical patterns of Section 3. Theorem 3 delivers the baseline disposition effect with no winning or losing experiences: stock returns are positively skewed, so the large returns in memory are predominantly gains, and the investor recalls, and sells, a gain. Theorem 4 shows that winning experiences decrease the tendency to sell gains; this is the attenuation in *Gain × Win Experience*. Theorem 5 shows that losing experiences increase the tendency to sell gains; this is the amplification in *Gain × Loss Experience*.

Further Implications

The theorems above were built to deliver the three patterns of Section 3. The same retrieval structure bears on four regularities documented elsewhere, which the model organizes rather than tests. Section 6 takes one prediction of the model to the data.

The first two concern which stocks get sold. Ben-David and Hirshleifer (2012) show that the probability of selling a position is V-shaped in its return since purchase: it rises with the size of gains and, at short holding periods, with the size of losses, with the gain arm steeper and little evidence of a jump at zero. They read the pattern as consistent with trading on beliefs revised by large price moves. In the model it follows from retrieval: large returns of either sign are the most retrievable (Corollary 1), so the most extreme positions are the ones that come to mind, and a mechanism that works through the size of the return implies no discontinuity at zero. Hartzmark (2015) shows that investors are more likely to sell the best- and worst-performing positions in their portfolios, holding each position’s own return fixed, and that firm-specific information does not explain the pattern. The model does not derive the rank effect: it sorts positions by the size of their returns, not by their standing within the portfolio. But retrieval in the model is competitive among the positions the cue reaches (Theorem 1), which is the structure a rank effect requires.

The third is evidence that what the investor has in mind at the moment of decision governs what she sells. Frydman and Rangel (2014) withhold the purchase price from the trading screen in an experiment, and the disposition effect falls; subjects still respond to the gain or loss they remember, but less. Frydman and Wang (2020) vary the same margin in the field, in the opposite direction: when a Chinese brokerage made the purchase price more prominent on its interface, the disposition effect rose by roughly 17%. Quispe-Torreblanca et al. (2026) show that the highest price a stock reached while the investor owned it acts as a second reference point for the decision to sell, and that peaks reached before she bought it do not. Quispe-Torreblanca et al. (2024) show that the gain or loss since the investor last logged in matters alongside the gain or loss since purchase, so that a login creates a new reference point. In each, the variation is in what the investor has in mind about the position, not in the position itself; the model takes that as the state at the moment of sale.

Holding period is a further instance of the third point, on what the investor has in mind at the moment of sale. Two facts are well established (Ben-David and Hirshleifer, 2012). First, the propensity to sell a position falls steeply with the time it has been held; combined with the disposition effect, this means that winners are sold before they age, so the positions that remain in a portfolio for a long time are disproportionately losers. Second, conditional on holding period, the gap between the propensity to sell a winner and the propensity to sell a loser is smaller for older positions. Both have a natural reading in retrieved-context theory, in which recently encoded episodes are more retrievable than older ones (Howard and Kahana, 2002). A recently acquired position comes to mind more readily than one held for a long time, and because the gain or loss on a recent position was encoded in a context close to the current one, with the same market conditions and the same news, its return category is more retrievable as well. The model of this section holds context fixed and so contains neither effect; contextual drift is the natural way to introduce both. Separating the resulting holding-period effects from the experience effects we document is not straightforward, since both accumulate with an investor’s time in the sample, and we leave a model of the two together to future work. Our claim is the narrower one: the model explains the experience asymmetry, not the holding-period schedule.

The fourth is diversification. Within the model the disposition effect feeds on the right tail of the return distribution: large gains dominate retrieval under the “stocks I own” cue because individual-stock returns are right-skewed. Diversification compresses that tail, so the model predicts a weaker disposition effect for mutual funds than for stocks. Chang, Solomon and Westerfield (2016a) show that the same investors exhibit a positive disposition effect in individual stocks and a negative one in mutual funds, and in experiments they trace the reversal to cognitive dissonance: investors avoid realizing losses they attribute to their own choices and readily realize losses they can attribute to a fund manager. The attenuation is what retrieval predicts; the reversal is not, and their account explains it. In the same experiments, showing subjects the reason they recorded when buying a stock raises the disposition effect for stocks; the reason is an added cue tied to the purchase episode.¹³

¹³Bailey, Kumar and Ng (2011), Ivković and Weisbenner (2009), and Calvet, Campbell and Sodini (2009) also find a weaker disposition effect for mutual funds and other delegated or diversified vehicles.

5 Alternative Mechanisms

We next examine alternative explanations for the asymmetric relationship between experience and selling. These include mechanical effects arising from the construction of experience measures, unobserved trading skill, the portfolio-level disposition effect (An et al., 2024), regression to the mean, recent returns, portfolio rebalancing, and trading induced by overconfidence.

5.1 Is the Experience Effect Mechanical?

Investors who have been trading for longer tend to accumulate more positions. If trading activity does not scale with positions, then the probability of selling any one position will fall with the number of positions, and thus also with trading experience. Because disposition is measured by taking the difference between the propensity to sell winners and the propensity to sell losers, the overall size of the effect might diminish with experience due to this effect. Furthermore, if investors who have been trading for longer tend to realize more winners, then the disposition effect would fall, mechanically, with winning experiences. To rule out this confound, we turn to simulations.¹⁴

We conduct a Monte Carlo simulation in which investors sell positions independently of their past experience and rerun our statistical analyses on the simulated data. We calibrate portfolio dynamics to observed moments in the data. First, as the reasoning in the previous paragraph suggests, within-investor experience and portfolio size are strongly positively correlated.¹⁵ It is roughly the same across the two types of experiences: +0.61 for winning experience, +0.58 for losing. We also calibrate to the decline in the held winner share as experience accumulates. Winning experience is negatively correlated with the held winner share (−0.26). The held winner share also declines as losing experience accumulates, but less strongly (the correlation is −0.16). Appendix A3 details the data-generating process and plots the simulated distributions. Because selling in the simulation does not depend

¹⁴A more obvious confound may appear to be that as the investor sells winners, there are fewer winners in the portfolio. Under random selling, this reduces the unconditional probability of a winner being sold. It does not, however, reduce the probability of sale conditional on being a winner, which is how the disposition effect is measured.

¹⁵All our measures are within-investor because our analyses employ investor fixed effects.

on experience, any nonzero interaction coefficient in the simulated data is mechanical by construction.

Table 6 reports results under three nulls: random selling, an experience-independent disposition effect, and an experience-independent disposition effect combined with the tendency to sell extreme positions documented by Ben-David and Hirshleifer (2012) (“V-shape selling behavior”). Under random selling, one is not more likely to sell a gain than a loss. Experience correlates with a greater number of positions, but in this simulation there is no baseline disposition effect, and so the propensity to sell a gain over a loss is zero and thus does not fall with experience. This explains the near-zero coefficients on the interaction between gain and both types of experiences.¹⁶ Random selling, combined with experience, cannot mechanically produce the effects we observe.

Our second null is a fixed disposition effect. Indeed, the simulation shows that experience mechanically diminishes this fixed disposition effect: losing and winning experiences lead to a lower propensity to sell winners for the reason we describe at the start of this section. This is potentially an issue for interpreting studies focusing on all experiences, rather than the differential effect of winning and losing experiences. Under our calibration, however, this mechanical effect is too small to explain the observed decline in the disposition effect with experience. Because winning experiences and losing experiences correlate with large portfolios about equally, the confound that we are concerned about, namely that winning experience indicates more experience overall, and hence a larger portfolio and a smaller disposition effect, cannot explain our results. The reason is that nearly the same dynamics work for losing experiences, resulting in a negative coefficient on the interaction, the opposite sign of what we find. A fixed disposition effect, combined with experience, also cannot mechanically produce the effects we observe.

Under the second null, the mechanical coefficients are small for both types of experience but less negative for winning experience than for losing experience. This ordering is opposite to the model prediction and the empirical estimates. As emphasized above, the

¹⁶The win-side coefficient under random selling, -0.0022 , is small economically, but too large to be simulation noise. Under equal sale probabilities the true winner-loser gap is zero by construction. As mentioned above, the propensity to sell any one position is decreasing in experience, and does so in a non-linear way. The regression, however, estimates a linear effect. The interaction terms between gain and winning and losing experiences pick up the residual from the misspecification.

main confound arises from portfolio size. A larger portfolio mechanically leads to a reduced disposition effect. A secondary mechanical effect offsets this main confound: as an investor sells winners, there will be fewer as time goes by, raising the probability of a sale under the fixed disposition effect. Winning experiences should thus increase the propensity to sell winners. This portfolio composition effect offsets the portfolio growth effect in our second null, causing the coefficient on the interaction between gain and winning experience to be closer to zero than that between gain and losing experience.

Our third null is a fixed disposition effect, combined with a tendency to sell the largest winners and losers that is not modulated by experience. One might think that a tendency to remove the most extreme positions would work roughly equally on the gain and the loss side. Each period, the extreme positions are removed, but there is an overall tendency to sell winners. As the number of positions grows, the disposition effect diminishes, though less for winners due to the portfolio composition effect described just above. Table 6 shows, however, that the portfolio composition effect is stronger than in the second null, so much so that the sign on the interaction between gain and winning experience is positive. There are two reasons why the portfolio composition effect is stronger. The first is that, due to skewness, extreme returns tend to be winners. The second is that winners tend to be removed earlier in the simulation for a given investor. The effect of removing winners matters more when there are fewer of them. The net effect is the opposite of what we observe in the data. On the loss side, the coefficient is essentially unchanged from the second null.

5.2 Skill and the Portfolio-Level Disposition Effect

An et al. (2024) document that the disposition effect is sharply lower when the investor’s portfolio is at a gain. We see a similar pattern in our data (Table 7, Base column). One explanation is skill. Skilled investors are more likely to hold portfolios at a gain, and skill may also reduce the disposition effect. Such an explanation suggests a possible confound. More realized gains would indicate more skill, and hence a smaller disposition effect, while more realized losses would indicate less skill, and hence a larger one. Because our experience measures count realized gains and losses, skill could in principle produce both interactions we document. Investor fixed effects do not rule this out, because skill, like experience, can

Table 6: Monte Carlo Assessment of Mechanical Bias ($|X| > 5\%$)

	Gain $\times$ LossExp	Gain $\times$ WinExp
<i>Paper estimates (Table 3, Column 5)</i>		
Coefficient	+0.0212	−0.0198
<i>t</i> -statistic	3.28	−3.30
<i>Scenario A: Random selling</i>		
MC mean	+0.0006	−0.0022
95% MC CI	[−0.020, +0.020]	[−0.024, +0.018]
<i>p</i> _{MC}	0.025	0.040
<i>Scenario B: Fixed DE</i>		
MC mean	−0.0064	−0.0032
95% MC CI	[−0.025, +0.014]	[−0.023, +0.015]
<i>p</i> _{MC}	0.005	0.045
<i>Scenario C: Fixed DE + V-shaped selling</i>		
MC mean	−0.0068	+0.0019
95% MC CI	[−0.025, +0.012]	[−0.013, +0.018]
<i>p</i> _{MC}	0.005	0.005

Notes. This table reports the Gain $\times$ Experience interaction coefficients from 200 Monte Carlo replications of the specification in Table 3, Column (5), at the $|X| > 5\%$ threshold; Figure 4 in Appendix A3 plots the corresponding distributions, where the 30%-threshold pattern is also discussed. Under all three scenarios, selling probabilities are independent of accumulated experience; any nonzero interaction coefficient is therefore a mechanical artifact of portfolio-composition changes. “MC mean” is the average coefficient across the 200 simulations; “95% MC CI” is the 2.5th–97.5th percentile interval across simulations; *p*_{MC} is the fraction of simulations producing a coefficient at least as extreme as the paper’s estimate in the direction of the paper’s finding. The simulation is calibrated to both within-investor co-movements that govern this channel: the correlation between experience and portfolio size (+0.61 winning, +0.58 losing) and the correlation between experience and the held winner share (−0.26 and −0.16).

accumulate.

We therefore condition on the state of the portfolio. Following An et al. (2024), we add an indicator for whether the portfolio is at a gain to the position-level specification, interacted with the position-level gain and with experience.¹⁷ The regression specification is

$$\begin{aligned}
\text{Sale}_{i,j,t} = & \alpha + \beta_1 \times \text{Gain}_{i,j,t} + \beta_2 \times \text{Portfolio Gain}_{i,t} + \beta_3 \times (\text{Portfolio Gain}_{i,t} \times \text{Gain}_{i,j,t}) \\
& + \beta_4 \times \text{Experience}_{i,t} + \beta_5 \times (\text{Gain}_{i,j,t} \times \text{Experience}_{i,t}) \\
& + \beta_6 \times (\text{Portfolio Gain}_{i,t} \times \text{Experience}_{i,t}) \\
& + \beta_7 \times (\text{Portfolio Gain}_{i,t} \times \text{Gain}_{i,j,t} \times \text{Experience}_{i,t}) + \epsilon_{i,j,t}
\end{aligned} \tag{15}$$

¹⁷We report results for the 30% threshold. Results are robust to the threshold definition.

where Portfolio Gain = 1 if the total unrealized return on the account is positive at the time of sale. Table 7 reports the estimates. With the *Portfolio Gain* $\times$ *Gain* interaction in the regression, the *Gain* coefficient (≈ 5.2) measures the disposition effect when the portfolio is at a loss. When the portfolio is at a gain, the disposition effect is the sum of β_1 and β_3 , which is negative in our sample. We thus reproduce the effect of An et al. (2024).¹⁸

We now turn to our main effect. The **Loss Exp.** and **Win Exp.** columns split experience into losing and winning trades. For portfolios at a gain, an additional loss experience adds $\beta_5 + \beta_7 = 0.2466$ percentage points to the disposition effect, whereas an additional win experience adds only 0.0393 (Panel B). Each underlying interaction is individually significant at the 1% level, but we do not report standard errors for the sums or a test of the difference between them. Having a portfolio at a gain could be correlated with experience, but if so, even experienced investors are influenced by whether these experiences are losses versus gains.

For losing portfolios, we do not find our effect in the baseline numbers: an additional experience of either type compresses the disposition effect by a similar amount ($\beta_5 = -0.0591$ for losing experience, -0.0663 for winning). Note, however, that the previous subsection shows that our effect has to push against the portfolio growth and portfolio composition effects. The portfolio composition effect is especially strong when there are few winners, as is the case when the portfolio is at a loss. We therefore caution against interpreting these results. The point estimates indicate that conditioning on whether the portfolio is at a gain does not eliminate the difference between winning and losing experience. This comparison does not, however, rule out time-varying investor skill.

5.3 Regression to the Mean

Because experience counts and the investor’s past measured disposition effect are both constructed from realized sales, regression to the mean could generate the estimated signs. Suppose every investor has the same true disposition effect and realized selling is noisy around

¹⁸An et al. (2024) interpret the pattern as mental accounting. Portfolio composition offers a further candidate: a portfolio at a gain typically holds many positions at small gains, so a tendency to sell winners is spread over many positions, and the probability that any one of them is sold is small. We do not take a stance on the source of this effect, which is beyond the scope of our paper.

Table 7: Experience-Driven and Portfolio-Driven Disposition Effect

Panel A: Experience-Driven and Portfolio-Based Disposition Effect				
Dependent Variable: <i>Experience variable in cols. (2)–(4):</i>	Sale $\times$ 100			
	Base	Total Exp.	Loss Exp.	Win Exp.
Gain	5.1834*** (10.01)	5.7315*** (9.83)	5.3401*** (9.89)	5.2954*** (9.96)
Portfolio Gain	2.7031*** (10.38)	3.1320*** (10.54)	2.8601*** (10.51)	2.8106*** (10.67)
Portfolio Gain $\times$ Gain	-7.9336*** (-13.26)	-8.6287*** (-13.52)	-8.3488*** (-13.48)	-8.1018*** (-13.30)
Experience		0.0035** (2.42)	0.1001*** (4.36)	0.0712 (0.82)
Gain $\times$ Experience		-0.0041*** (-3.42)	-0.0591*** (-3.50)	-0.0663*** (-3.08)
Portfolio Gain $\times$ Experience		-0.0036*** (-3.48)	-0.1215*** (-3.08)	-0.0773*** (-6.03)
Portfolio Gain $\times$ Experience $\times$ Gain		0.0053*** (4.33)	0.3057*** (3.15)	0.1056*** (3.87)
Constant	8.5824*** (46.99)	8.1990*** (32.09)	8.4401*** (43.79)	8.5357*** (38.79)
Observations	43,649,867	43,649,867	43,649,867	43,649,867
R-squared	0.141	0.142	0.142	0.142

Panel B: Experience and the Disposition Effect in the Gain State		
	Loss Experience	Win Experience
Effect on the winner–loser gap ($\beta_5 + \beta_7$)	0.2466	0.0393

Notes. This table presents estimates from the linear probability model described in equation 15, where the dependent variable is a dummy equal to 1 if a stock is sold, multiplied by 100 for ease of interpretation. Panel A examines how individual trading experience and portfolio-level performance jointly shape the disposition effect. The four columns differ in which experience variable enters the regression as “Experience”: Column (1) includes baseline controls only; Column (2) uses *Total Experience* – the cumulative number of prior trades; Column (3) uses *Loss Experience* – the cumulative count of prior trades with a realized return below -30% ; Column (4) uses *Win Experience* – the cumulative count of prior trades with a realized return above $+30\%$. Row labels “Experience”, “Gain $\times$ Experience”, etc. therefore refer to whichever variable is indicated in the column header.

Gain is an indicator for whether the stock has an unrealized capital gain at the time of decision. *Portfolio Gain* is a dummy indicating whether the investor’s portfolio is, on net, in a gain position. The interaction terms *Portfolio Gain* $\times$ *Gain* and its higher-order interactions capture how broader portfolio context moderates the disposition effect.

Panel B reports the effect of one additional experience on the winner–loser gap in sale propensity when the portfolio is at a gain, computed as $\beta_5 + \beta_7$ from the corresponding column of Panel A; statistical significance is not reported for these combined effects.

All regressions include investor fixed effects, stock fixed effects, and time fixed effects. Standard errors are clustered by investor, time, and stock. The sample period is July 2013 to February 2016. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

it. An investor who has realized many winners is then, by construction, one whose past measured disposition effect was high by chance, and regression to the mean predicts a lower measured effect going forward; one who has realized many losers had a low past draw and should revert upward. Both interactions would follow with no change in behavior at all.¹⁹

Under this explanation, the relationship between experience and the disposition effect should weaken as longer histories average out noise in past selling decisions. We re-estimate the baseline specification on the subsample of investors with account tenure above the sample median. Table 8 reports the results against the full-sample headline from Table 3, Column (5).

The data move the opposite way. Both interactions strengthen in the long-history subsample: $Gain \times Loss Experience$ rises from +0.0212 ($t = 3.28$) to +0.0302 ($t = 4.37$), and $Gain \times Win Experience$ moves from -0.0198 ($t = -3.30$) to -0.0219 ($t = -5.31$), both significant at the 1% level. Regression to the mean predicts attenuation where the noise in past measured behavior has had the most chance to average out; the response to accumulated outcomes instead proves durable. Account tenure in this sample varies over months, not decades (interquartile range 266–362 days, Table 1), so this split gauges the stability of the effect within short histories; it does not measure the size of an investor’s lifetime store of competing memories, the margin the age test in Section 6 addresses.

5.4 Extrapolation and Recent Returns

Recent stock returns may be correlated with realized gains and with investors’ selling decisions. If investors extrapolate these returns, the negative interaction between winning experience and $Gain$ could reflect return expectations rather than recall of past realized outcomes.

To separate the two channels, we re-estimate the full 5% specification adding the held stock’s own recent return as a control, measured over the prior week and, separately, the prior month. Table 9 shows the control leaves the experience interactions intact. With the prior-week return, $Gain \times Loss Experience$ is +0.0218 ($t = 3.37$) and $Gain \times Win Experience$ is -0.0199 ($t = -3.35$); with the prior-month return they are +0.0216 ($t = 3.36$) and -0.0200

¹⁹We thank Stefano Cassella for raising this possibility.

Table 8: Regression to the Mean: The Long-History Subsample (5% Threshold)

Dependent Variable:	Sale $\times$ 100	
	(1) Full sample	(2) Long-history sub-sample
Gain	2.4781*** (4.35)	1.5401*** (2.64)
Loss Experience	0.0109 (1.13)	-0.0101** (-2.33)
Win Experience	0.0034 (0.36)	0.0291*** (5.00)
Gain $\times$ Loss Experience	+0.0212*** (3.28)	+0.0302*** (4.37)
Gain $\times$ Win Experience	-0.0198*** (-3.30)	-0.0219*** (-5.31)
Constant	8.4974*** (36.63)	7.7038*** (35.75)
Observations	43,649,867	28,734,001
R-squared	0.138	0.137

Notes. Column (1) reproduces Column (5) of Table 3 (the full-sample headline at the $|X| > 5\%$ threshold). Column (2) re-estimates the same specification on the subsample of investors with account tenure above the sample median. The dependent variable is an indicator for sale, multiplied by 100. *Loss Experience* and *Win Experience* count prior trades with realized returns below -5% or above $+5\%$, respectively, relative to the investor's average purchase price. *Gain* equals 1 if the position has an unrealized gain at the decision time. Both columns include individual, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t -statistics are in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively. If the interactions reflected regression to the mean in noisy measured behavior, they should attenuate in the long-history subsample, where that noise has had more chance to average out; instead, both interactions strengthen.

($t = -3.36$), against the headline $+0.0212$ and -0.0198 . The recent-return coefficients are positive (weekly 5.50, $t = 1.75$; monthly 4.04, $t = 3.04$). These coefficients indicate a positive association between recent returns and selling; they do not by themselves distinguish return extrapolation from the V-shaped selling pattern documented by Ben-David and Hirshleifer (2012). Controlling for recent returns leaves the experience interactions largely unchanged, but does not distinguish recall from all belief-based explanations. Enke, Schwerter and Zimmermann (2024) show experimentally that cued recall of past news can generate overreaction in beliefs and prices. Recall could therefore affect selling through expectations as well as through the retrieval of past experiences. Distinguishing these channels would require belief measures that our data do not contain.

Table 9: The Experience-Driven Disposition Effect, Controlling for Recent Returns (5% Threshold)

Dependent Variable:	Sale $\times$ 100		
	(1) Headline	(2) + Past-week	(3) + Past-month
Gain	2.4781*** (4.35)	2.2190*** (3.80)	2.1403*** (3.70)
Loss Experience	0.0109 (1.13)	0.0112 (1.17)	0.0116 (1.21)
Win Experience	0.0034 (0.36)	0.0033 (0.34)	0.0030 (0.32)
Gain $\times$ Loss Experience	0.0212*** (3.28)	0.0218*** (3.37)	0.0216*** (3.36)
Gain $\times$ Win Experience	-0.0198*** (-3.30)	-0.0199*** (-3.35)	-0.0200*** (-3.36)
Past-week return		5.4957* (1.75)	
Past-month return			4.0379*** (3.04)
Constant	8.4974*** (36.63)	8.3418*** (36.05)	8.3255*** (36.23)
Observations	43,649,867	42,526,853	42,526,853
R-squared	0.138	0.138	0.138

Notes. Column (1) reproduces the full specification from Table 3, Column (5). Columns (2) and (3) add the held stock's cumulative return over the prior week and the prior month, respectively, as a control. *Loss Experience* and *Win Experience* count prior trades with realized returns below -5% or above $+5\%$, respectively, relative to the average purchase price. *Gain* equals 1 if the position has an unrealized gain at the decision time. All columns include investor, stock, and time fixed effects, and standard errors are clustered by investor, time, and stock. Columns (2)–(3) have fewer observations because positions without a full recent-return history are dropped. The sample period is July 2013 to February 2016. *t*-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

5.5 Portfolio Rebalancing

Selling winners need not reflect any bias. When a holding appreciates, its weight in the portfolio rises above the investor’s target, and a risk-managing investor trims it; a holding that falls becomes underweight and is held. A disposition effect—and the experience interactions we document—could be the footprint of rational rebalancing.

Under a rebalancing rule based only on current portfolio weights and fixed target weights, purchase prices have no direct role in the sale decision. Our *Gain* indicator depends on the purchase price, but our baseline does not control explicitly for current portfolio weights and therefore does not compare positions with identical weights. Evidence from [Frydman and Wang \(2020\)](#) supports a role for the purchase-price reference point: making that price more prominent on a Chinese brokerage’s trading interface increased the disposition effect by roughly 17%. This intervention changed the information displayed to investors without directly changing portfolio weights.

A rebalancing rule based on current weights and fixed targets does not depend directly on whether past trades realized gains or losses. Our estimates show that the winner–loser sale gap increases with losing experience ($Gain \times Loss\ Experience > 0$) and decreases with winning experience ($Gain \times Win\ Experience < 0$). Past outcomes may nevertheless be correlated with current weights or changes in target weights. Investor, stock, and time fixed effects do not eliminate these time-varying relationships, so history independence alone does not rule out rebalancing.

Three further observations bear on this explanation. Our results hold when partial sales are excluded and only complete liquidations are retained (Section 2.3). This restricts the role of incremental rebalancing ([Odean, 1998](#)), although full liquidation may still reflect a change in target weights. The direction of the bias also runs against tax-motivated selling, which favors realizing losses over gains; throughout our sample, individuals’ realized gains on A-shares acquired on the secondary market were exempt from individual income tax, so neither loss-harvesting nor gain-deferral can produce the pattern.²⁰ [Chang, Solomon and](#)

²⁰The exemption is the temporary one in force since Caishuizi [1998] No. 61 (Ministry of Finance and State Administration of Taxation); only transfers of restricted shares are taxed (Caishui [2009] No. 167), which does not apply to the secondary-market trades in our sample. Sell-side trading costs remained: a 0.1% stamp duty levied on the seller only, and dividend taxes differentiated by holding period under Caishui

Westerfield (2016a) document a positive disposition effect in individual stocks and a negative one in mutual funds held by the same investors. An explanation based on rebalancing would need to account for this difference across asset classes.

Our baseline omits a position’s current portfolio weight, so *Gain* may partly proxy for an overweight position. Controlling for the held stock’s recent return leaves the experience interactions essentially unchanged (Table 9), but recent returns do not fully measure portfolio weights.

5.6 Trading Intensity and Overconfidence

Realized gains may increase investor overconfidence and trading activity (Barber and Odean, 2000b; Statman, Thorley and Vorkink, 2006). If the additional sales are less sensitive to whether a position is at a gain or loss, higher turnover could reduce the measured disposition effect. This explanation predicts a positive relationship between winning experience and turnover.

We regress each investor’s monthly turnover—value bought and sold during the month, scaled by twice average portfolio value—on the cumulative count of winning experiences, with investor fixed effects and standard errors clustered by investor. The sign goes the wrong way for overconfidence. Winning experience lowers turnover (Table 10: -0.29 , $t = -17.8$ with investor fixed effects): investors who have realized more gains trade less, not more. The negative coefficient rejects the specific overconfidence channel in which realized gains dilute the measured disposition effect through heavier trading; the recall model itself makes no prediction for total turnover, which bundles purchases, loser sales, and portfolio growth.

6 Testing the Model

The model predicts weaker responses to realized outcomes when more past episodes compete for recall (Theorem 2). In the proof of Theorem 4, each winning experience lowers the ratio of recalling large to small returns by an amount inversely proportional to the number of small-

[2012] No. 85, with dividends on shares held over one year exempt from September 2015 (Caishui [2015] No. 101).

Table 10: Winning Experience and Monthly Turnover

Dependent Variable:	Monthly Turnover	
	(1)	(2)
Win Experience	-0.0825*** (-10.14)	-0.2911*** (-17.83)
Constant	7.3127*** (116.51)	8.4999*** (82.97)
Observations	881,576	838,399
R-squared	0.001	0.313
Investor FE	No	Yes

Notes. The dependent variable is investor-level monthly turnover, defined as total purchase plus sale value during the month divided by twice the investor’s average portfolio value, $(\text{buy} + \text{sell}) / (2 \times \text{average portfolio value})$. *Win Experience* is the cumulative count of an investor’s prior realized gains exceeding 5%. Column (1) excludes and Column (2) includes investor fixed effects. Standard errors are clustered by investor; *t*-statistics in parentheses. The unit of observation is the investor-month; the sample period is July 2013 to February 2016. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

return episodes among the stocks the investor owns, and each losing experience raises that ratio by an amount that likewise shrinks as the number of such episodes grows (Theorem 5). An investor with a short trading history has few such episodes, so a realized gain or loss should move her selling more than the same outcome moves an investor with a long history. Age is our proxy for the number of such episodes, the store of memories that compete for recall. The logic is the one [Malmendier and Nagel \(2011\)](#) apply to beliefs, that a recent outcome is a larger share of a young person’s experience and so moves her more; here the share is of the episodes that compete for recall rather than of years lived. [Korniotis and Kumar \(2011\)](#) document a different effect of age, the erosion of investment skill by cognitive aging, which concerns the quality of decisions rather than the response to outcomes.

We test the prediction by re-estimating the full specification separately for investors at or below and above the median age of 36. Table 11 reports the estimates. Both interactions are more than twice as large for younger investors: the $\text{Gain} \times \text{Win Experience}$ coefficient is -0.0354 for younger investors against -0.0154 for older ones, and $\text{Gain} \times \text{Loss Experience}$ is $+0.0402$ against $+0.0160$; all four are significant with the predicted sign. The baseline *Gain* coefficients are 2.30 and 2.59 in the younger and older groups, respectively. The experience interaction estimates are larger in absolute value among younger investors, consistent with the model’s prediction that an additional outcome has a greater effect when fewer past episodes compete for recall.

Table 11: Experience-Driven Disposition Effect by Age Group (5% Threshold)

Dependent Variable: $ X > 5\%$	Sale $\times$ 100			
	Age ≤ 36		Age > 36	
	(1) Main	(2) Full	(3) Main	(4) Full
Gain	2.3935*** (4.20)	2.2976*** (3.28)	2.5789*** (5.20)	2.5874*** (4.91)
Loss Experience	0.0172*** (3.92)	0.0037 (0.72)	0.0180* (1.85)	0.0124 (1.11)
Gain $\times$ Loss Experience		0.0402*** (3.62)		0.0160*** (2.66)
Win Experience	0.0329*** (6.69)	0.0425*** (7.44)	-0.0070 (-0.78)	-0.0026 (-0.25)
Gain $\times$ Win Experience		-0.0354*** (-5.91)		-0.0154** (-2.55)
Constant	8.3169*** (40.31)	8.4253*** (34.95)	8.1570*** (37.11)	8.1998*** (36.68)
Observations	19,396,148	19,396,148	24,253,718	24,253,718
R-squared	0.145	0.145	0.133	0.133

Notes. This table re-estimates the disposition-effect regression separately for younger (age ≤ 36) and older (age > 36) investors, split at the median investor age of 36. Columns (1) and (3) include the *Loss Experience* and *Win Experience* main effects only; Columns (2) and (4) add both *Gain* $\times$ experience interactions, matching the full specification of Table 3, Column (5). *Loss Experience* and *Win Experience* count prior trades with realized returns below -5% or above $+5\%$, respectively, relative to the average purchase price. *Gain* equals 1 if the position has an unrealized gain at the decision time. All specifications include investor, stock, and time fixed effects, and standard errors are clustered by investor, time, and stock. The sample period is July 2013 to February 2016. *t*-statistics are in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

7 Conclusion

Using transaction records for 189,530 Chinese retail investors over the 2013–2016 boom and bust, we show that the content of past trading experience—not merely its accumulation—governs how the disposition effect evolves. The bias does not simply fade with activity: realized gains weaken it, while realized losses sharpen it. A two-standard-deviation increase in accumulated large losses raises the propensity to sell winners over losers by about 23% of the baseline disposition effect, an equivalent increase in gains lowers it by about 21%, and the calibrated simulations do not reproduce the positive loss-side interaction.

Our cued-recall model explains the baseline disposition effect and its response to experience. Each trade is stored as an episode—its return, the action taken, and the surrounding context—and the cue “stocks I own” retrieves past episodes in proportion to their similarity and frequency, each competing with the others for recall. Because large gains occur more often than large losses, gain memories surface more readily and generate the baseline bias even for novices; realizing a gain then re-tags it as disposed rather than owned, muting its later pull on recall, whereas realizing a loss reinforces it. A single retrieval channel reproduces the baseline effect, its decay after gains, and its amplification after losses, with no shift in preferences and no reference-dependent utility parameters.

The findings suggest that trading experience need not reduce the disposition effect uniformly. Studies conducted mainly in rising markets may place greater weight on winning experience. The estimated experience interactions are larger in absolute value among younger investors, consistent with the model’s prediction that an additional outcome has a greater effect when it represents a larger share of past experience.

Two implications follow. First, winning experience is associated with lower turnover, contrary to the prediction that realized gains increase trading through overconfidence (Barber and Odean, 2000b; Statman, Thorley and Vorkink, 2006). The smaller winner–loser sale gap is consistent with the model’s prediction that realizing a gain weakens its association with ownership. Second, Gödker, Jiao and Smeets (2025) show that selectively recalled gains shape beliefs, and Enke, Schwerter and Zimmermann (2024) show that cued recall makes expectations overreact; pairing transaction records with elicited beliefs would test directly how remembered outcomes move return expectations.

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Appendix A Additional Information

A1 Calculating Trading Returns

Transaction-Level Returns We follow the return methodology of [Hartzmark \(2015\)](#) and [Frydman, Hartzmark and Solomon \(2018\)](#): the realized return on a sale is the difference between the sale price and the average purchase price of the position. When additional shares are acquired after the initial purchase, we measure the return against the volume-weighted average of all purchase prices for the position.

For each buy order b and sell order s of each stock j at time t , the realized return on a sale is

$$return(s, j)_t = \frac{vol(s, j)_t \times (price(s, j)_t - price(\bar{b}, j)_t)}{cost(b, j)_t} \quad (16)$$

where $vol(s, j)_t$ is the volume of stock j sold at time t , $price(s, j)_t$ is the selling price, $price(\bar{b}, j)_t$ is the average purchase price of stock j up to time t , and $cost(b, j)_t$ is the cumulative cost of purchasing stock j up to time t :

$$price(\bar{b}, j)_t = \frac{holding_{j,t-1} \times price(\bar{b}, j)_{t-1} + price(b, j)_t \times vol(b, j)_t}{holding_{j,t-1} + vol(b, j)_t} \quad (17)$$

$$holding_{j,t} = \sum_{t=0}^t [vol(b, j)_t - vol(s, j)_t] \quad (18)$$

$$cost(b, j)_t = \sum_{t=0}^t [vol(b, j)_t \times price(b, j)_t] \quad (19)$$

For example, suppose five transactions comprise three buy orders and two sell orders. The sell orders leave $cost(b, j)_t$ unchanged, since $vol(b, j)_t = 0$ on those dates; only the buy orders move it. The five transactions yield two realized returns, one per sell order. A stock is treated as new if an investor, having previously held shares in it, sells the entire position before buying again.

This procedure yields a return for every sale. The same average-purchase-price reference also defines a paper return for every position an investor continues to hold. Paper returns, though more often negative than positive, are strongly right-skewed (skewness ≈ 23)—the input the recall model of [Section 4](#) acts on. Realized returns are far more symmetric, since investors close positions from across the distribution.

Stock-Level Returns We also compute a total return at the stock level, from the first purchase to the final sale that liquidates the holding:

$$return_j = \frac{\sum_{t=0}^t [vol(s, j)_t \times price(s, j)_t]}{\sum_{t=0}^t [vol(b, j)_t \times price(b, j)_t]} - 1 \quad (20)$$

This is the net gain from trading stock j relative to its total purchase cost.

Other Approaches Barrot, Kaniel and Sraer (2016) consolidate trades on a per-individual, per-stock, and per-day basis, sourcing daily stock returns from EUROFIDAI. This approach simplifies the computation of stock returns, enabling the subsequent analysis to regress daily abnormal returns for each stock.

A2 Empirical Strategies for Quantifying Investor Experience

We construct trading episodes following Seru, Shumway and Stoffman (2010). Each investor’s history is segmented into episodes as follows:

1. Initiation of a Trading Episode:

- An episode begins when the investor purchases shares of a stock.

2. Termination of a Trading Episode:

- An episode ends at the first sale, partial or complete, of those shares.
- The realized return at that sale classifies the episode as a winning or losing experience.

3. Implications of Partial Sales:

- A partial sale ends the current episode.
- The remaining shares begin a new episode from that point.

These episodes define the experience counts used to relate past realized outcomes to current selling decisions.

A3 Mechanical Bias from Portfolio Composition

Our experience measures count an investor’s own past realized gains and losses, so they are built from the same portfolio process that generates the current selling decision. Two features of that process could reach the estimated interactions mechanically. The first is scale: as experience accumulates the book grows, and the probability of selling any one position falls with the number of positions, compressing the winner–loser sale gap on both sides. The second is composition: realized trades thin the book, and realizing gains leaves fewer winners. Under random selling, fewer winners lower the unconditional probability that the next sale is a winner, but not the probability of sale conditional on being a winner, which is how the disposition effect is measured. With a baseline disposition effect, however, there is less competition among the remaining winners, and the probability of sale conditional on being a winner rises, pushing the $\text{Gain} \times \text{WinExp}$ interaction upward, against our negative estimate. The same arithmetic applied to realized losses thins the losers and pushes the $\text{Gain} \times \text{LossExp}$ interaction upward, toward our positive estimate. This appendix measures the net effect of these channels.

Three features of the design and the data bound this channel.

Portfolios evolve between sales. The composition channel requires that the thinning of winners or losers at t persists to the sell decision at $t+1$. In practice, both prices and holdings evolve substantially between consecutive trades. Price movements reclassify positions from gains to losses (and vice versa), while new purchases add stocks whose gain/loss status is unrelated to the stock sold at t . In our data, the median interval between consecutive sales by the same investor is approximately 12 trading days, during which the typical stock experiences price changes large enough to alter its gain/loss classification. Consequently, a sale at time t need not systematically shift the winner–loser composition at $t+1$.

Identification is within-investor. Our main specification (equation 3) includes investor fixed effects, which absorb any time-invariant tendency of an investor’s portfolio to be tilted toward gains or losses. The identifying variation comes from within-investor changes in experience interacted with within-investor changes in the gain indicator. For the mechanical channel to bias the experience–gain interaction coefficients, it would need to generate a systematic within-investor correlation between the cumulative count of past large gains (losses) and the conditional probability of the next sale being a winner, net of investor-level means. In our data this within-investor co-movement is substantial: the within-investor correlation between portfolio size and experience is +0.61 for winning experience and +0.58 for losing, and the correlation between the held winner share and experience is -0.26 and -0.16 , re-

spectively. We calibrate the simulation below to reproduce these correlations and measure the resulting mechanical bias.

A calibrated simulation. To quantify the mechanical channel directly, we conduct a Monte Carlo simulation described in detail below. We simulate 5,000 investors whose number of trades is drawn from a right-skewed distribution (mean ≈ 27), and we calibrate portfolio sizes, experience counts, and—critically—both within-investor co-movements that drive the mechanical channel to the data: the correlation between experience and portfolio size (+0.61 winning, +0.58 losing) and the correlation between experience and the held winner share (−0.26 and −0.16; untabulated calibration moments). Under three null scenarios—random selling, a fixed (experience-independent) disposition effect, and a fixed disposition effect combined with a V-shaped selling schedule—we estimate the interaction specification of Table 3, Column (5), adapted to the simulated panel; any nonzero interaction is, by construction, mechanical.

Monte Carlo Assessment of Mechanical Bias

This subsection details the simulation. Selling is generated under null rules independent of past experience, so that any nonzero Gain $\times$ Experience coefficient is, by construction, a mechanical artifact of portfolio composition.

Simulation design. We simulate 5,000 investors and repeat the exercise 200 times. The number of sell decisions per investor is drawn from a log-normal distribution (median ≈ 9 , mean ≈ 27 , with a heavy right tail capped at 600), reproducing the dispersion and skew of the experience counts in Table 1; a fixed count per investor cannot match these moments. Each investor starts with about five stocks and, after each sale, buys back toward a size target that drifts upward over the trading sequence. This growth is the feature that lets the simulated panel reproduce both the mean portfolio size (7.6) and the substantial within-investor correlation between experience and portfolio size in the data—+0.61 for winning experience and +0.58 for losing; a constant target produces a within-investor correlation near zero. The held winner share, in turn, falls as experience accumulates (−0.26 and −0.16 in the data): we reproduce this with a bounded decline in the market level over each investor’s trading sequence. Indexing the decline to progress through that sequence limits the total price decline and avoids an excessively heavy left tail in simulated returns. Between sell decisions, prices evolve over approximately five trading days as a log-normal process with two independent jump components—frequent moderate jumps and rare large jumps (crashes and spikes)—calibrated to the mean counts of winning and losing experiences at the 5% and

30% thresholds in Table 1. Algorithm 1 summarizes the data-generating process for a single investor.

Algorithm 1. Data-generating process for one investor

1. Draw the number of sales $n = \text{round}(\text{LogNormal}(2.20, 1.48))$, capped to $[1, 600]$.
2. Draw the starting size $k_0 = \max(3, \text{round}(5 + \mathcal{N}(0, 0.6^2)))$; open $K = k_0$ positions at the purchase price. Draw the investor’s market-decline amplitude a_i (mean 0.08).
3. For each sell event $\tau = 1, \dots, n$:
 - (a) If $K < 3$, replenish to k_0 positions.
 - (b) Evolve prices for ≈ 5 days: $p \leftarrow p \cdot \exp(\mathcal{N}(\mu, \sigma^2) + J_{\text{mod}}^+ + J_{\text{big}}^+ + J_{\text{mod}}^- + J_{\text{big}}^-)$, with independent moderate and rare-large jumps.
 - (c) Scale prices to the bounded market level $M(\tau) = \exp(-a_i(\tau - 1)/(n - 1))$, applied incrementally relative to $\tau - 1$. Tying the level to sequence progress caps the total move at a_i , so the held winner share declines with experience.
 - (d) Classify each holding as a gain or loss from its unrealized return.
 - (e) Assign sell weights—**(A)** equal; **(B)** $1.5\times$ for gains; **(C)** rule (B) $\times (1 + 0.8|r|/0.10)$ —and sell one position with probability proportional to its weight.
 - (f) Record all current holdings: the sale indicator, the gain dummy, and the running counts WinExp and LossExp.
 - (g) If the realized return exceeds $\pm 5\%$ (resp. $\pm 30\%$), increment the matching experience counter.
 - (h) Remove the sold position; set the target $= \min(k_0 + \min(0.22(\tau - 1), 2.5), 40)$ and stochastically buy back toward it.

We consider three null scenarios:

- (A) **Random selling.** Each stock in the portfolio is equally likely to be sold, regardless of whether it is at a gain or loss. This isolates the pure portfolio-composition channel.
- (B) **Fixed disposition effect.** Winners are sold with $1.5\times$ the probability of losers, but this ratio is constant across all experience levels. This tests whether a standard (experience-independent) disposition effect, combined with the mechanical channel, can generate the estimated interaction terms.
- (C) **Fixed DE with V-shaped selling.** Following Ben-David and Hirshleifer (2012), we additionally weight stocks with extreme absolute returns more heavily in the sell decision, while keeping the selling rule independent of accumulated experience. This is the most conservative null.

For each simulated dataset, we estimate the Column (5) specification of Table 3 adapted to the simulated panel:

$$\begin{aligned} \text{Sale}_{i,j,t} \times 100 = & \alpha_i + \beta_1 \text{Gain}_{i,j,t} + \beta_2 \text{LossExp}_{i,t} + \beta_3 \text{WinExp}_{i,t} \\ & + \beta_4 \text{Gain}_{i,j,t} \times \text{LossExp}_{i,t} + \beta_5 \text{Gain}_{i,j,t} \times \text{WinExp}_{i,t} + \varepsilon_{i,j,t} \end{aligned}$$

with investor fixed effects and standard errors clustered by investor. The empirical specification adds stock and time fixed effects with three-way clustering; the simulated panel has no meaningful stock or calendar dimension, so those terms would absorb nothing by construction. The coefficients of interest are β_4 and β_5 . Under all three null scenarios, any nonzero estimate is purely mechanical.

Results. The main-text Table 6 reports the mean simulated coefficient, its 95% Monte Carlo confidence interval, and the Monte Carlo p -value across 200 replications; Figure 4 plots the corresponding distributions at the 5% threshold. The one-sided p_{MC} gives the fraction of simulations producing a coefficient at least as extreme as the empirical estimate in the direction of the paper’s finding.

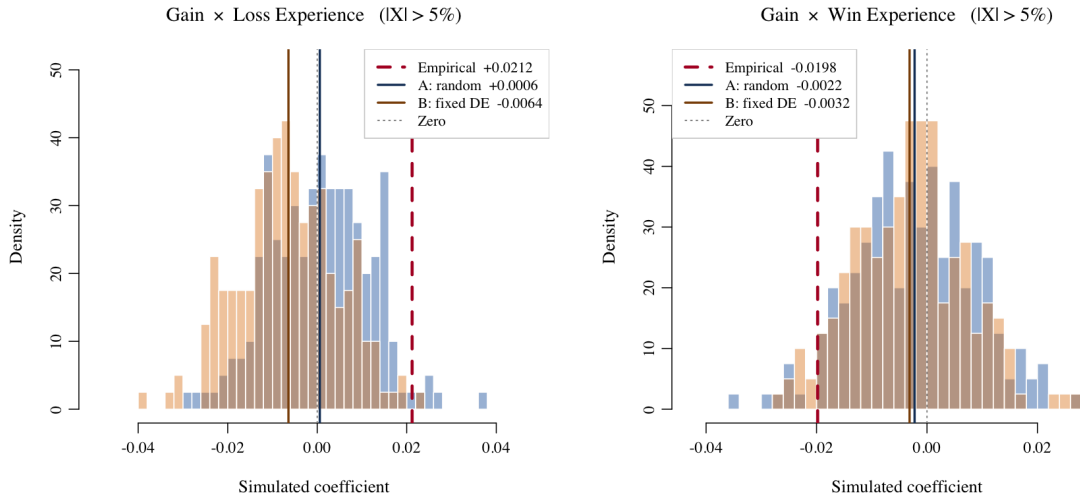


Figure 4: Distribution of Simulated Gain $\times$ Experience Coefficients ($|X| > 5\%$)

Notes. The two panels show the distribution of the Gain $\times$ Loss Experience (left) and Gain $\times$ Win Experience (right) coefficients across 200 Monte Carlo replications at the $|X| > 5\%$ threshold, under the data-matched calibration (within-investor experience–portfolio-size correlation +0.61 and experience–winner-share correlation -0.26). Blue histograms correspond to Scenario A (random selling); orange histograms to Scenario B (fixed disposition effect); Scenario C (fixed DE + V-shape) is reported in Table 6 but not plotted here. Dashed red lines mark the paper’s point estimates from Table 3. Both mechanical distributions are centered near zero—the loss-side slightly negative, opposite the empirical +0.0212, and the win-side near zero, far short of the empirical -0.0198 —so each empirical estimate falls well into the tail of what portfolio arithmetic alone produces.

Interpretation. At the 5% threshold, the mean loss-side interaction under each calibrated null is negative or close to zero. The mean $\text{Gain} \times \text{LossExp}$ coefficient is -0.0064 under the fixed disposition effect and -0.0068 under the V-shaped selling rule motivated by Ben-David and Hirshleifer (2012), compared with the empirical estimate of $+0.0212$ ($t = 3.28$). Under random selling, the mean coefficient is close to zero ($+0.0006$). In no scenario does more than one simulation in forty ($p_{\text{MC}} \leq 0.025$) reach the empirical value. These results indicate that the calibrated mechanical effects do not explain the positive loss-side estimate.

For winning experience, the simulations incorporate two offsetting relationships observed within investors. Portfolio size increases with experience (correlation $+0.61$), which makes the mechanical $\text{Gain} \times \text{WinExp}$ coefficient more negative. The held winner share decreases with experience (correlation -0.26), which makes the coefficient more positive. The resulting mean coefficients are -0.0032 under Scenario B and $+0.0019$ under Scenario C, compared with the empirical estimate of -0.0198 . Fewer than one simulation in twenty ($p_{\text{MC}} \leq 0.045$) reaches the empirical value. Matching only the portfolio-size correlation would yield a mechanical bias of roughly half the empirical estimate; matching both relationships produces a substantially smaller bias.

Under random selling, the winning-experience interaction is -0.0022 , even though the true winner–loser sale gap is zero at every experience level. This small interaction arises from the linear specification. Per-position sale probabilities decline nonlinearly with experience, with a steeper decline in small portfolios. Because the winner share also falls with experience, gain observations are concentrated where this decline is steeper, producing a more negative fitted slope. All three null distributions incorporate this specification effect.

At the 30% threshold, the mechanical loss-side interaction remains negative, whereas the empirical estimate is positive ($+0.0907$). The empirical winning-experience estimate is less precise (-0.0634 , $t = -1.79$), and the wide simulated distribution prevents a conclusion from that comparison.

At the 5% threshold, both empirical interactions are statistically distinguishable from the three calibrated nulls ($p_{\text{MC}} \leq 0.045$). The simulated loss-side interactions range from -0.0068 to $+0.0006$, and the negative mean winning-experience interaction accounts for at most one-sixth of the empirical estimate. These comparisons reject the specific mechanical explanations considered here; they do not establish a lower bound on a behavioral effect or identify recall as its source.